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Chemistry ST

Applied chemistry (Pokhara University)

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NEPAL ENGINEERING COLLEGE

CHEMISTRY MANUAL

FOR UNDERGRADUATE ENGINEERING STUDENTS

PROF. SABITRI K. TRIPATHI, Assist. Prof. Prabin Basnet , Assist. Prof. Sulochana Pradhan

10/2/2019

This manual is prepared based on Pokhara University Syllabus for undergraduate engineering students

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PREFACE

Undergraduate students of engineering have to study Chemistry as basic science course in the first or second semester for 4 credit hours. 4 lectures and 1 tutorial classes are designed for a semester.This manual is prepared targeting undergraduate engineering students especially under Pokhara University. Topics involved in this manual is according to Pokhara University syllabus. It covers all the topics according to the syllabus. The manual is prepared focusing all levels of students. The language is simple and easy, examples are relevant with clear explanations. Some structures and figures are dragged from google.

The manual is trying to justify the practical aspects of Chemistry to the engineering students at undergraduate level to develop their interest towards the subject. YouTube- links copied from google, are also inserted as a tool for easy understandings and learnings. Objectives and applications of each unit is clearly listed.In addition to this, at the end of each unit, there are list of questions so that students can practice.

There are total 7 Units. Unit3, 6 and 7 is written by Prof. Sabitri Tripathi, Unit 1 and 4 by Assist. Prof. Sulochana Pradhan and Unit2 and 4 is written by Assist. Prof. PrabinBasnet.

The Authors

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CHEMISTRY SYLLABUS OF POKHARA UNIVERSITY

CHM 103.4 Chemistry (4-1-2)

	Theory	Practical	Total
Sessional	30	20	50
Final	50	-	50
Total	80	20	100

Course objectives :

1. Analyze chemical behavior of materials
2. Analyze water quality
3. Analyze environmental aspects of various elements and compounds

Unit **Content** **Hours**

1. Ionic Equilibria and Electrochemistry 10
   Introduction and types of buffer solutions, mechanism, Hinderson-Hassel Balch Equation, electrochemical cells, Galvanic cells, cell notation, cell reaction, cell potential, single electrode potential, standard electode potential, electrochemical series, and its applications, Nernst equation, corrosion and its type, mechanism and control
2. General Inorganic Chemistry 10
   Ionization energy, electro negativity, electron affinity, characteristic properties of S and P block elements, introduction of transition metals, characteristic properties of transition mentals(electronic configuration, atomic radii, variable oxidation states, complex formation, color and magnetic properties)
3. General Organic Chemistry 10
   Reaction Intermediates: carbocations, carbanions, carbon free radicals, Stereoisomerism, Organic reaction mechanism: Nucleophilic substitution reaction(SN1&SN2), Electrophilic aromatic substitution reaction, Elimination (E1&E2)reactions, Electrophilic and free radical addition reactions
4. Polymer Chemistry 10
   Polymer and polymerization, basic concept, types of polymerization, (addition and condensation), thermoplastics and thermosetting plastics, preparation properties and uses of : polyethylene, PVC, Teflon, bakelites, nylon, polyester, polyurethane and silicon, rubber, processing of natural rubber and vulcanization
5. Analytical Chemistry 6
   Introduction and application of following analytical techniques: Fractional distillation, chromatography (paper and TLC), NMR and Mass spectroscopy
6. Industrial Chemistry 4
   Introduction of paints, chemistry of paints, lubricants and its classification, cement, chemistry of cement, manufacture and its setting mechanism, explosives: TNT, TNG
7. Environmental Chemistry 10
   Water pollution: causes of water pollution, acid rain, alkalinity, COD, DO, Hardness(effects to human health),control measures

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Air pollution: causes, global warming and climate change, ozone layer depletion, and CFC, control measures

Soil pollution: cause, effects and its control measures

Laboratory Works

1. Determination of total alkalinity of given water sample
2. Determination of total hardness of given water sample
3. Determination of fre chlorine in the given water sample
4. Preparation of buffer solutions and determination of their pH value
5. Estimation of DO in the given water sample
6. Estimation of COD in the given water sample
7. To separate ink mixture by paper chromatography or TLC(Demo)
8. To purify sample of a mixture of crude alcohol and petroleum by fractional distillation(Demo)
9. To estimate carbon monoxide gas in the car exhaust

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Why Chemistry for Engineering Students?

Engineering is the application of science and mathematics to solve technical problems and create new systems, products or devices to benefit civilization. The Accreditation Board for Engineering and Technology (ABET), is a professional organization that oversees engineering education. According to ABET, engineering is defined as “the profession in which a knowledge of the mathematical and natural sciences gained by study, experience and practice is applied with judgment to develop ways to utilize, economically, the materials and forces of nature for the benefit of mankind”. Therefore, engineering requires applied science, and chemistry is the center of all sciences. The more chemistry an engineer’s understand, better global problem will be addressed.

A real engineer should be able to do anything. All types of Engineers play with materials, for example, civil engineers work with concrete, cement, metals etc, computer engineers with semi conductors, data Electronic and communication engineers with semi conductors, different types of metals and nano materials, data, Electrical engineers with circuits and environment is the cross cutting issues for all branches of engineering.

Chemistry is what we see and perceive in the macroscopic world is the result of interaction at atomic level. It helps us to understand the most important issues day to day around us. For example: need of clean water, how climate change, how chemical energy in fossil fuel/solar power is converted into useable mechanical and electrical energy for our cars and homes and how chemical fertilizers are manufactured to boost food security at our farm. The knowledge gained through chemistry allows us to make informed decisions for our future. A strong chemistry curriculum should provide the opportunity to the students to solve real world problem and convey to others. Therefore, the main objectives of engineering curriculum is not to learn to be an engineer, but to get the knowledge as broad as possible which might be useful at any point in their professional life.

This manual is bridging the concepts and theory of chemistry with examples from fields of practical application, thus reinforcing the connection between science and engineering. It deals with the basic principles of various branches of chemistry, namely, physical chemistry, inorganic chemistry, organic chemistry, analytical chemistry, polymer chemistry, cement chemistry, environmental and industrial chemistry. And all kinds of engineers need to understand the materials we work with every day - their physical and chemical properties

Finally, Chemistry is everywhere; engineering would be weaker without it. Chemistry deals with **‘WHY’** and Engineering deals with **“WHAT”.** Both should go together to solve real world problem!!!

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Table of Content

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UNIT3: GENERAL ORGANIC CHEMISTRY

WRITTEN BY: **PROF. SABITRI K. TRIPATHI**

Objective :
Students are able to know about the behavior of organic molecules which is applicable in their real /professional life

Applications

- Study of Human, plant and animal metabolism/cytology etc.
- Industry
- Water and Environment
- Farming and food production etc.

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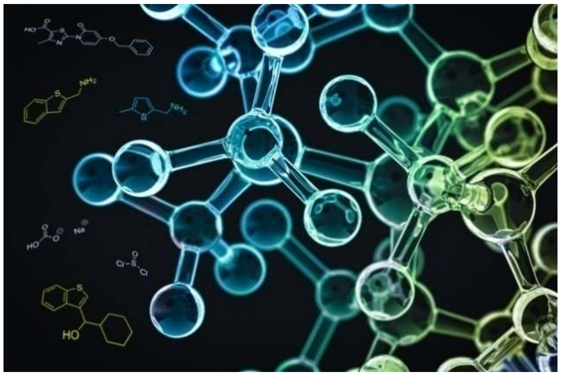
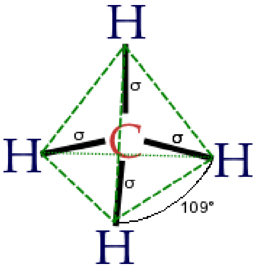
ORGANIC REACTION MECHANISM

The hydrocarbons and their derivatives are known as organic compounds. These are found in nature or synthesized at laboratories. There are more than 9million organic compounds known where as inorganic compounds is very small number. Our body, food, medicine, daily appliances, all is made up of organic compounds. Therefore, it is necessary to understand the behavior of organic molecules and their reactions. Generally, inorganic reactions are simple and fast whereas organic reactions are complicated and involve different steps to complete. It is important to know how the reaction completes involving different steps which is known as reaction mechanism to understand completely the behavior of molecules in different environments.





Structure of Methane: Simplest organic molecule

All organic compounds necessarily consist of carbon and hydrogen atom. They have covalent bonding. Carbon has the property that it can bond with numerous number of carbon atom forming long chain (even billions of atoms). This property of carbon is known as Catenation. Similarly carbon atom can form double and triple bonds which further makes interesting to understand its behavior. This is the region that there is large number of organic compounds possible.

Electron shifting process during organic reaction
All organic compounds are formed by covalent bond which results due to sharing of electrons. While there is organic reaction, there must be bond breaking and new bond formation process. Bond breaking means shifting of electron/ or electron pairs. Shifting of electrons is possible due to some forces which are:

- Inductive effect
- Mesomeric effect
- Resonance effect
- Hyperconjugation effect

Bond fissionprocess:
There are two types of bond fission process which are:
<u>Homolytic bond fission :</u> When the shared pair of electrons in covalent bonding is shifted equally to the bonded group after breaking of the bond, there is no loss or gain of electrons by the group. They acquired their own electron; hence no polarity development occurs instead both bear's free electrons of their own. It means after cleavage they have equal distribution of electron and hence this type of fission is known as homolytic bond fission. Free radicals can be generated by homolytic fission. This reaction has wide

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application in polymer industry, themal plants, radioactive emissions etc. Ozone depletion is one of the causes of free radical reaction due to CFC molecule.

Examples: CH_3 : Cl ⟶ °CH_3 + Cl°
Methyl free radical Chlorine free radical

Heterolyticbondfission: If there is unequal distribution of electron/s after cleavage of covalent bond, there is difference in electron density, polarity is developed in the corresponding groups. The group which gains electron acquire negative charge and the group which losses its electron, becomes positively charge. It means there is hetero distribution of electrons after bond cleavage. Carbocations and carbanions can be generated by this way. For example:
Examples: CH_3: Cl ⟶ +CH_3 + :Cl-

Methyl carbocation Chloride ion
CH_3: X ⟶ -CH_3 + :X+
Methyl Carbanion

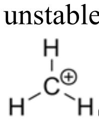
REACTION INTERMEDIATES

Generally, organic reactions are slow and thus the reaction procedures can be tracked. In the organic reactions, before the formation of final product, some intermediate step/s are involved with the formation of unstable species. They are known as reaction intermediates. They bear (+ve or -ve) charge or sometimes having free electrons and therefore they are highly unstable. There are different types of reaction intermediates which are:

1. Carbocations
2. Carbanions
3. Carbon free radicals

1. Carbocations

Carbon is tetravalent. It has 4 bonding state. Let us take the simplest molecule, methane. Carbon is bonded with 4-hydrogen atoms. If the 4^{th} hydrogen atom is lost by taking the shared pair of electrons, carbon has develop a positive charge, such carbon with positive center is known as carbocation (carbon with cation). It is a reaction intermediate since it bears positive charge which makes it unstable.


Carbocation

Methods of generation of carbocations:

1. By direct ionization:
   R-X ⟶ R^+ + X^-
2. By addition of positive species in the unsaturated system
   $CH_3CH{=}CH_2$ +H+ ⟶ $CH_3HC^+CH_3$

Classification

Carbocations can be classified as primary, secondary and tertiary based on the number of alkyl groups directly attached with the positively charged carbon.

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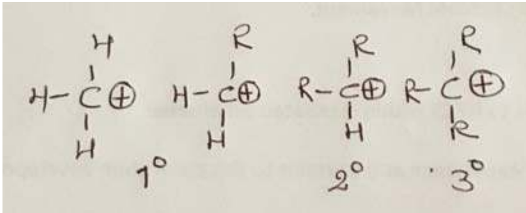
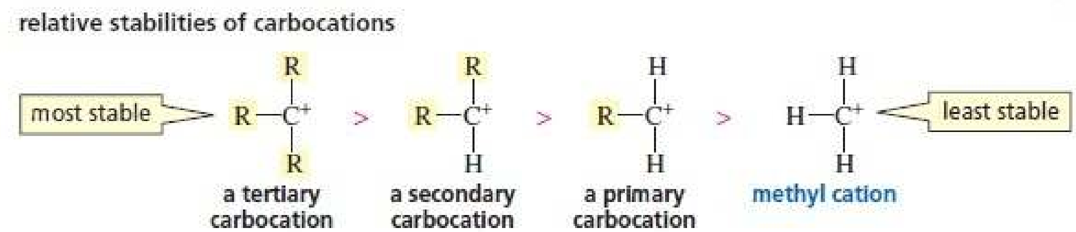


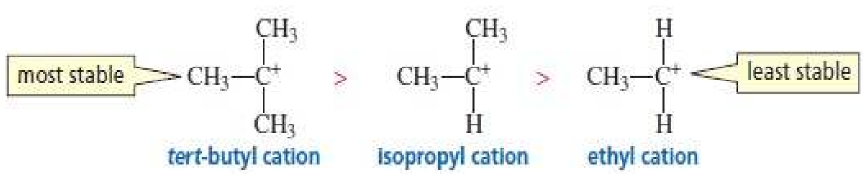
Stabilities of Carbocations:
All carbocations are highly unstable. However, there instabilities are still different from each other because of different numbers of alkyl groups attached with it.   Stabilities of carbocations can be explained on the following basis:

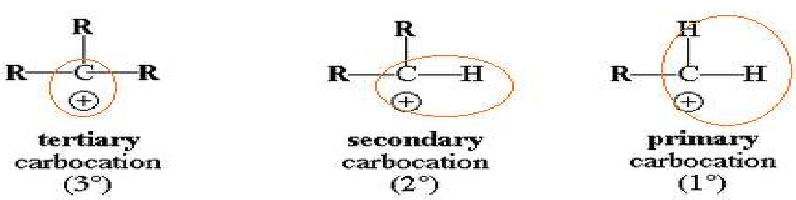
- Inductive effect: In case of tertiary carbocation, there are three alkyl groups present. It pushes its electrons towards positively charged central carbon via +Inductive effect, which causes to minimize the magnitude of positive charge. Since there are 3-alkyl groups, +I-effect stabilizes the carbocation most. In case of secondary and primary carbocation, +I effect causing groups are less, so less effect of decreasing the magnitude of positive chare. In this way the stability order is as follows:



- Hyperconjugation effect: More the delocalization of positive charge, greater will be the stability and vice versa. This can be explained on the basis of hyperconjugative effect. In case of tertiary carbocation, 9 hyperconjugative structures can be obtained. It means, the positive charge will be delocalized most.In case of secondary carbocation, six hyperconjugative structures can be obtained where as in primary only three. So the stability order is:

$$3° > 2° > 1° > {}^{+}CH_3$$





More the +I group, more will be the stability of the carbocation

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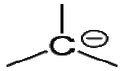
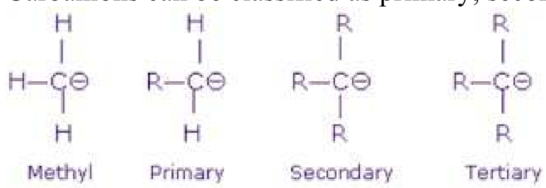
2.Carbanion 

Carbanions are reaction intermediates in which carbon possess negative charge . There is total 8electrons, 6 are bonding and two electrons remain free for bonding. It means it acquires one extra electron due to which it bears a negative charge. It is generated by heterolytic bond fission. Thus its hybridisation is sp3 and geometry is pryamidal.

Generation methods:

1. By ionization:
   R-H $\longrightarrow$ R- +H+
2. When negative ions add to carbon carbon double or triple bond
   $H_2C{=}CH_2$ + Y- $\longrightarrow$ H_2C- - CH_2Y
3. Due to resonance
   CO_2^- R− + CO_2

Classification:
Carbanions can be classified as primary, secondary and tertiary.

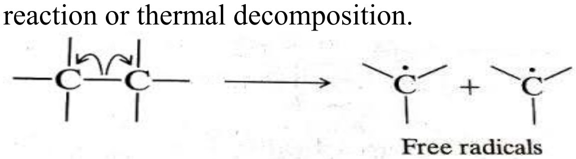


Stabilities
Like carbocations, stabilities of carbanions can be explained on the basis of Inductive effect. The three alkyl groups present in tertiary carbanion helps to increase the magnitude of negative charge via +Inductive effect, so makes it most unstable. In case of secondary carbanion, only two alkyl groups are present whereas in case of primary carbanion, one alkyl group is present. Similarly in methyl carbanion, there is absence of alkyl group. As a result methyl carbanion is most stable carbanion. The stability order is therefore,

$$\Theta CH3 > 1° > 2° > 3°$$

3.Carbon Free Radicals:
Carbon free radicals are defined as carbon having odd or free electron. These results due to homolytic covalent bond fission. It is denoted by dot (°) against carbon atom at the top. They are electrically neutral and highly unstable. They readily try to pair up the odd electrons. There are 3pairs of bonded electrons and one unpaired free electron , so total number of electrons it bear is 7.

Generation methods:
Carbon free radicals can be generated by hemolytic bond fission which can be carried out by photolytic reaction or thermal decomposition.



1. By protonation of Alkenes or alcohols:
   $H_2C = CH_2 \leftrightarrow {}^{\circ}CH_2CH_3$
2. By decomposition of diazo compounds
   $C_2H_5N_2Cl \leftrightarrow {}^{\circ}C_6H_5 + N_2$

Classification:

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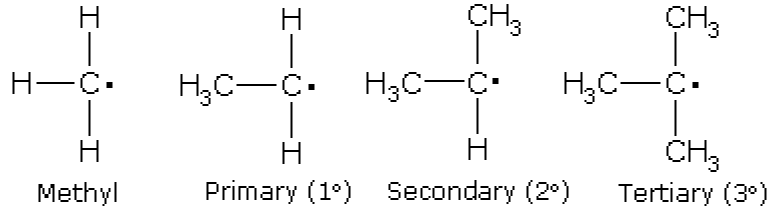
Carbon free radicals are classified as primary, secondary and tertiary on the basis of number of alkyl groups attached with it.



Stabilities:

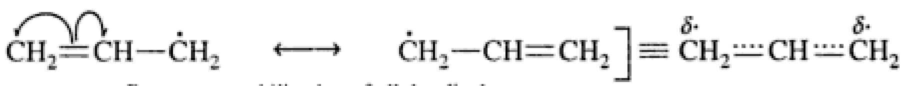
The stabilities of free radicals can be explained on the basis of:

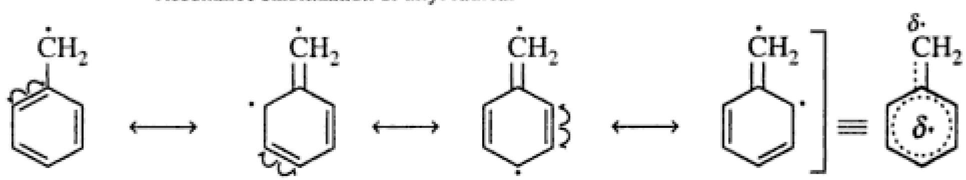
1. Inductive effect:

Free radicals are electron deficient group, seeking electron for its complete octet. The alkyl group has electron pushing effect, which is known as +Inductive effect. In case of tertiary carbon free radical, three alkyl groups are present which pushes their electrons partially towards carbon bearing free electron. As a result, electron density towards the central carbon with free electron increases. There are three alkyl groups which helps to make it stabilize. In case of secondary free radical, two alkyl groups are present which helps to stabilizes it where as for ethyl free radical, only one alkyl group is present and in methyl free radical, no alkyl group is present. Therefore,tertiary is most stable and methyl free radical is least stable.

2. Resonance effect:

Resonance effect can also explain the stability of carbon free radicals when the molecule bears double bond. For example in case of Benzyl free radical and Allyl free radical.


Resonance stabilization of allyl radical


Resonance stabilization of benzyl radical

The resonance stabilizes free radicals. More the resonance structure, greater the stability and vice versa.

3. Hyper conjugative effect:

Hyperconjugation is due to delocalization of sigma bond. Greater the hyperconjugative effect, higher the stability and vice versa.In case of tertiary carbon free radical, 9 hyperconjugative structures can be obtained, which means more delocalization of electrons, cause to stabilize the free radical most. In secondary carbon free radical, 7 hyperconjugative structures can be obtained where as in Ethyl only one .Therefore, the stability order is

Stability order is : 3°>2°>1°>°CH3

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STEREOISOMERISM

Stereochemistry:

The word stereoisomers is derived from Greek word, stereo -meaning towards space and isomers-meaning forms. So stereoisomers are the organic compounds having same molecular formula but the arrangement of atoms or group of atoms on carbon in three diamentional space is different.Lets take the Cabde compound which has following two spatial arrangements:

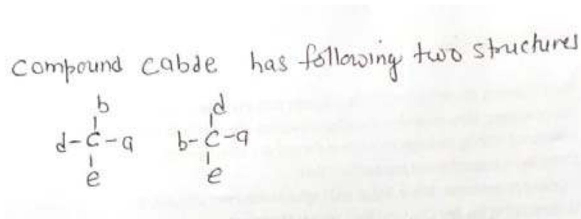
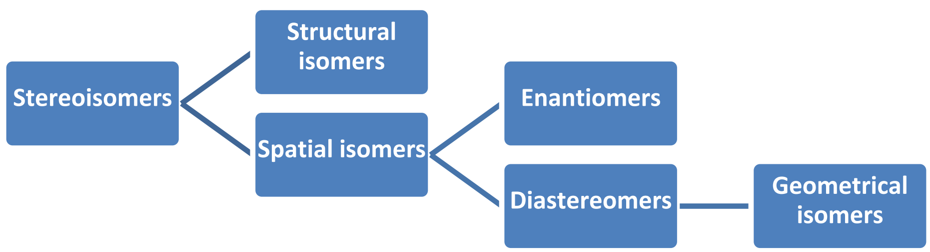


Illustration of Stereoisomerism

Both have same molecular formula Cabde, but having different spatial arrangements, therefore, they are stereoisomers of each other.

Classification:



Classification of Stereoisomers

Fundamentals of Stereochemistry:

Plane polarized light: Normal light has the character that it can rotate in all possible direction. When it is passed through nicoloid prism, the character of light changes and it has the property to rotate in only one plane. Such light which can vibrate in only one plane is known as **Plane polarized light**.

Optical activity and optically active substances: Organic substances which has the ability to rotate the plane of plane polarized light is known as optically active substance. The rotation may be two types. When rotation is clockwise, it is called dextro rotation(D) or/(+) rotation. Similarly, if the rotation is anti clockwise, it is called Laevo rotation or (L) or (-) rotation. The compound which are optically active can only shows optical isomerism. Enantiomers and Diasteriomers are the optical isomers. Optical activity can be tested by an instrument called Polarimeter as shown below:

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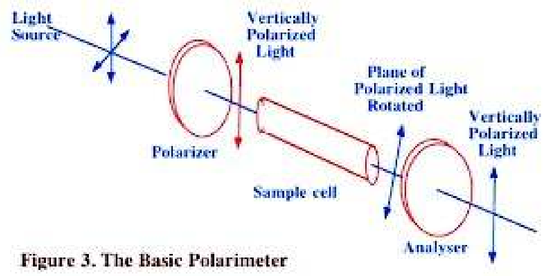
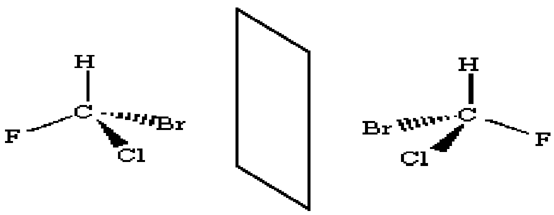


Figure 3. The Basic Polarimeter

Chiralcarbon : Carbon having four different groups attached is known as chiral carbon. Such carbon has no plane of symmetry and shows optical isomerism.
For example:

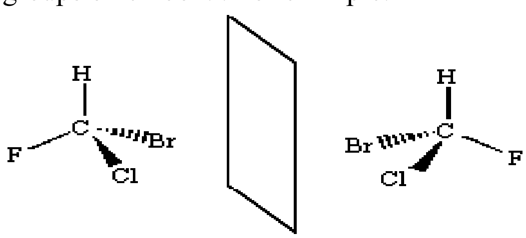


Bromochloro
flouromethane(object) Plane mirror Image

Optical Isomerism are enantiomers and diastereomers. These are optically active compounds. If one isomer rotate plane polarized light clockwise another must rotate anticlockwise.

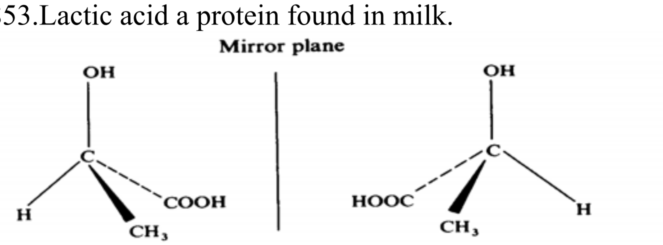
Enantiomers

Enantiomers are the optical isomers. They are non superimposable mirrior image stereoisomers. It means their mirror images cannot exactly coincide(superimpose) without altering the position of the attached groups on carbon. . For example:



Bromochloro
flouromethane(object) Plane mirror Image

Bromochloroflouro methane when kept in front of plane mirror, the images looks exactly as shown above. It means Bromo group appear closer where as flouro group looks farther. The image cannot superimpose. Hence they are enantiomers of each other. If object rotates PPL clockwise, image should rotate anti clockwise. Lactic acid is the first molecule in which optical isomerism was detected by Louis Pasteur in 1853.Lactic acid a protein found in milk.



Lactic acid (+) Lactic acid (-)

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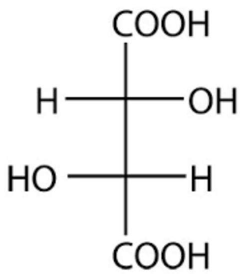
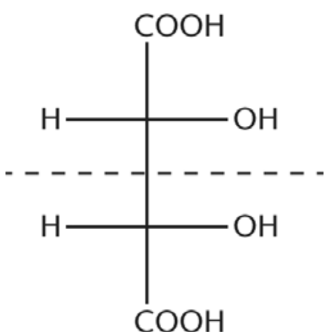
Condition for Enantiomerism:

Carbon must be chiral for enantiomerism. It means carbon should have four different linkages. However, if there are two or more than two chiral carbons, it is not necessary that the compound shows enantiomerism. The compound may have plane of symmetry. If there is plane of symmetry in the molecule, it is optically inactive. Therefore, for enantiomerism molecule must be dissymmetric, There should not be any plane of symmetry. For example: Tartaric acid has two carbon, chiral. It has different isomeric forms. In the figure below, Tartaric acid(I) is optically active but (II) is optically inactive. It is because, in case of (II), there is plane of symmetry (shown by dot lines in the structure, middle, horizontally). The upper half is mirror image of lower half. As a result, net rotation is zero and hence the molecule is optically inactive and known as meso compounds. Therefore, **chirality is necessary but not sufficient condition for enantiomerism.**



Tartaric acid (I): Optically active



Tartaric acid (II): optically inactive

Properties of Enantiomers:

Physical properties: Physical properties such as mp, bp, specific gravity are mostly identical. They have different rotation towards plane polarized light. If one rotates clockwise, another should have anti clockwise rotation.

Chemical properties:Chemical properties may be similar or sometimes different. For example, dextrorotatory glucose upon consumption, reacts inside our body, producing energy where as Laevo rotatory glucose do not have such action inside our body.

Diasereomers

These are molecules which are neither superimposable nor mirror image stereoisomers. They are optically active. For example:1,2 dihydroxy pentane molecule, 4- isomeric structures can be obtained as follows. I&II are mirror image of each other so they are enantiomeric pairs. Again, III&IV are enantiomers of each other. When we compare the isomers I&III, I&IV, II&III and II&IV, they are neither superimposable nor mirror images of each other. Such stereoisomers are known as Diastereomers.

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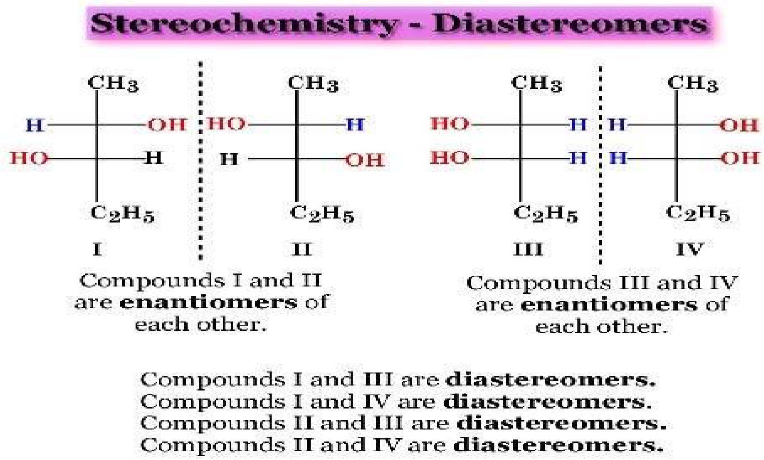
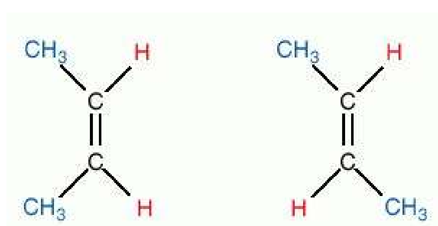


<u>Properties</u>

Physical properties: Diastereomers have different physical properties.
Chemical properties: They have often different chemical properties.

Geometrical Isomers:

In case of carbon carbon single bond system, there is one sigma bond between carbon and carbon, which allows free rotation to the group attached to it. In the double bond carbon system, there is one sigma and one pi bond. Pi bond is formed due to lateral overlap of p-orbitals. So it restricts the rotation of groups attached to it which causes two different orientation (spatial arrangement)of groups attached to carbon and carbon atom. Thus such structures which can be obtained by the restriction of rotation of carbon carbon double bond is known as geometrical isomers often known as cis-trans isomers. If same group are in same side of double bonded carbon, it is called cis isomers and if in opposite side, it is called as trans isomers.



cis-2-Butene *trans*-2-Butene

Note: In case of carbon carbon triple bond, rotation is restricted, but since in this system, only one group is attached in each carbon, there is no possibility of geometrical isomerism.

Necessary conditions for geometrical isomerism:

- There must be carbon carbon double bond
- Number one carbon has two different groups attached
- Number two carbon consist of same group as number one carbon
- If same group are in same side, it is called as cis isomers and if same groups are in opposite side, it is called as trans isomers
- If number two carbon has only one group same to number one carbon, the same group decide whether it is cis or trans isomer. For example:

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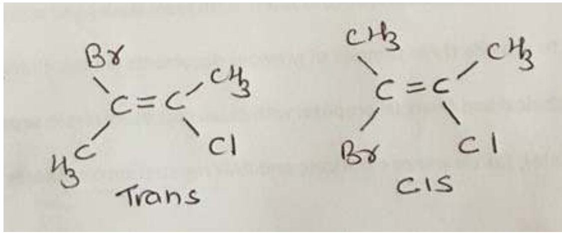
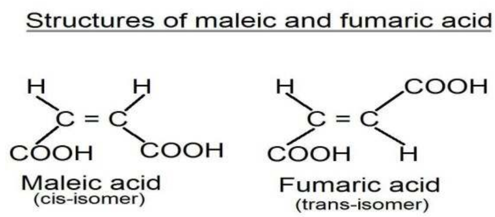


Bromo chloro butene



Maleic and Fumaric acids are very important while discussing geometrical isomerism. They are same molecules, with different sparial arrangement, exhibiting cis- trans isomerism. Therefore, they have different names.

Some Questions:

1. Some more examples of the compounds exhibiting geometrical isomerism are: ClBrC=CBrCl, HBrC=CBrH, H-C-OH= H -C- OH etc.Cis and trans isomers have different chemical and physical properties.**Some questions** *Ho*w stereoisomers differe from structural isomers?
2. What do you mean by enatiomers? What are the necessary conditions for enantiomerism?
3. Why lactic acid shows enantiomerism where as methane molecule does not?
4. Which are chiral molecules: Methanes, chloromethane, chlorobromo methane, lactic acid, chlorobromoiodo methane, ?
5. What do you mean by diasteriomers?
6. Which type of molecules show diastereomerism?
7. How enantiomers differe from diastereomers?
8. Why 1.2 dichloro pentane molecule shows both enentiomerism and diasteriomerism, explain it.
9. Write the necessary condition for geometrical isomerism
10. Only double bonded molecule shows cis trans isomerism, justify it.

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THE ORGANIC NAME REACTIONS AND THEIR MECHANISMS

Reagents for organic reactions:
Chemical substances due to which organic reaction is possible, is known as reagents. They are mainly two types:
Nucleophiles or Nucleophilic reagents
A nucleophile is a species (an ion or a molecule) which is strongly attracted to a region of positive charge either on carbon or any other positively charged ion. They are either fully negative ions, or else have a strongly -vecharge somewhere on a molecule, or sometimes neutral. Common nucleophiles are hydroxide ion(OH-), cyanide ion(CN-), water and ammonia molecules.
Electrophiles or Electrophilic reagents
Electrophiles are chemical substance, generally positively charged species, ion or molecule, which loves to attack towards –ve center of carbon or any other groups. For example: Nitronium ion(NO2+), hydronium ion(H3O+), Lewis acids, SO3 molecule.

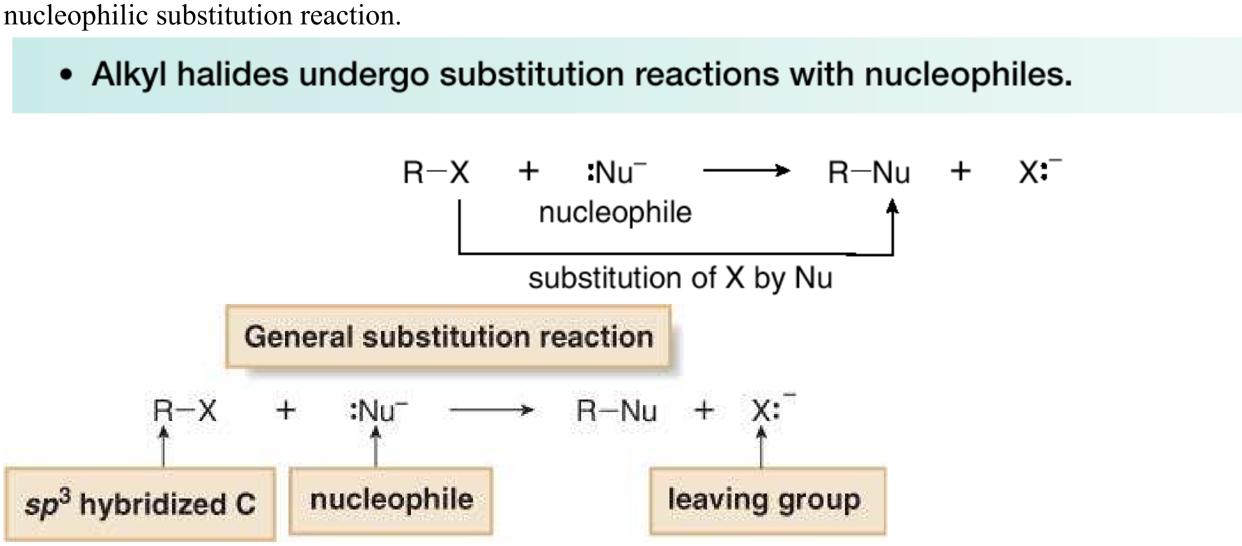
Types of organic reactions
- Substitution reaction
- Elimination reaction
- Addition reaction
- Rearrangement reaction

Substitution Reaction
When one species is displaced by another during reaction, without the change in nature of bond in the substrate molecule, the reaction is known as substitution reaction. For example:

$$AC + BD \rightarrow AD + BC$$

The substitution reaction can be divided into Nucleophilic and Electrophilic substitution reaction based on the reagents used.

Nucleophilic Substitution Reaction
When substitution reaction proceeds due to the presence of nucleophilic reagents, the reaction is known as nucleophilic substitution reaction.



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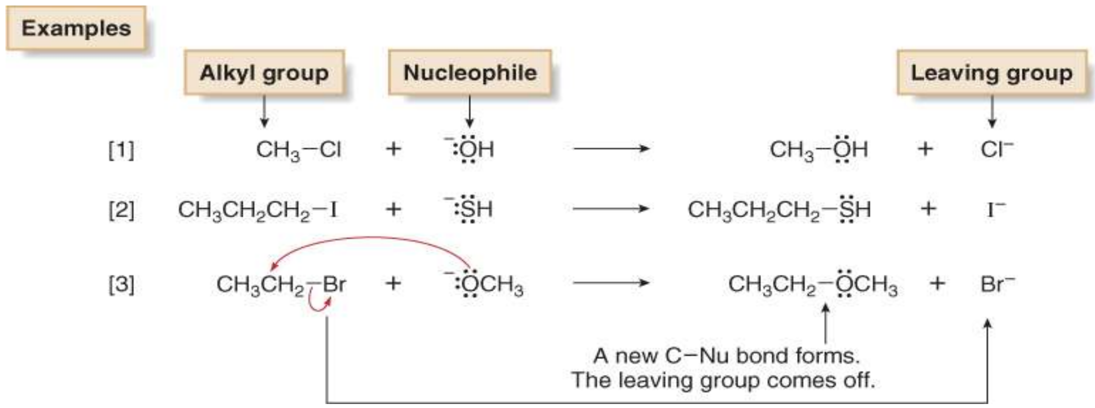


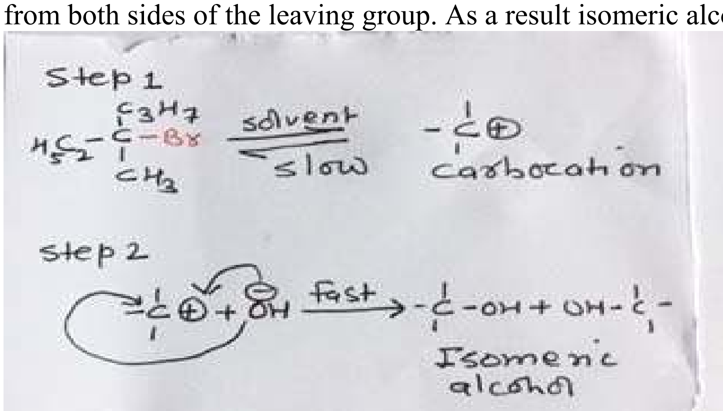
This reaction is further divided into two types:
SN1 Reaction: Nucleophilc substitution reaction by first order or unimolecular reaction
SN2 Reaction: Nucleophilic substitution reaction by second order or bimolecular reaction

SN1 reaction:
The nucleophilic substitution reaction when completes slowly involving two steps to complete, the reaction is known as SN1 reaction. When tertiary butyl bromide reacts with sodium hydroxide in presence of suitable solvent, a slow reaction leads to the formation of alcohol as final product. This is an example of SN1 reaction.

<u>Mechanism of SN1 Reaction</u>
Let's take the example of hydrolysis of tertiary butyl bromide. The whole reaction completes in two steps:

Step 1
This is a slow ionization step. The tertiary heptylbromide molecule in presence of ethanol or methanol as solvent molecule undergoes slow ionization. Leaving group, bromine which is a weak base and highly electronegative group. In presence of solvent molecule, Br group takes away the shared pair of electrons toward itself forming bromide ion, consequently, positively charged carbocation is generated as an intermediate product, which is highly reactive.

Step2
Once carbocation is generated, the nucleophile OH^- generated from KOH/NaOH attacks the carbocation from both sides of the leaving group. As a result isomeric alcohol is formed.



Actually, nucleophile, OH- starts to attack the carbocation once positive charge starts to develop on carbon which is virtually in the first step of the reaction as it is slow step. Once the leaving group

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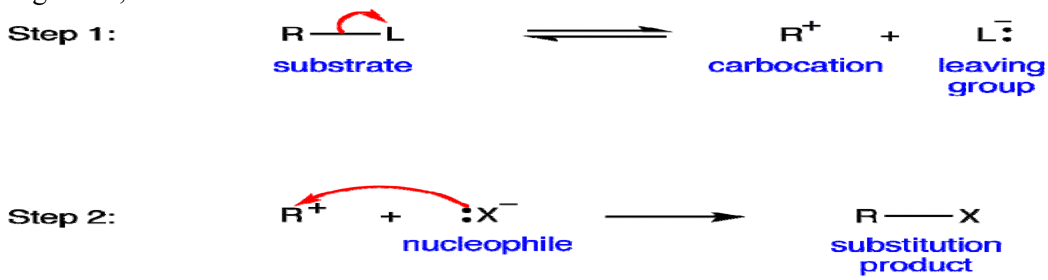
completely detaches, the nucleophile starts to attack from the side of leaving group. In this way two side attacks is possible.
In general,



<u>Kinetics of SN1 reaction:</u>
When the reaction is fast and rate depends on only one molecule, it is known as first order or unimolecular reaction. Rate of the molecule depends on slowest step of the reaction. If only one molecule is involved in the slowest step, that single molecule is responsible for the rate of overall reaction. In SN1 reaction, step 1 is slowest where only one molecule, i.e. substrate molecule is involved, so this reaction is known as first order or unimolecular reaction.

Mathematically,
Rate $\propto$ [Substratemolecule]
It means for SN1 reaction, only substrate molecule determines the rate of reaction regardless of the concentration of nucleophile molecule.

<u>Stereochemistry of SN1 reaction:</u>
SN1 reaction completes in two steps. First step is slow and second step is fast. Nucleophile attacks the carbocation by both sides of leaving group, as a result isomeric alcohol is formed as final product. They are of equal ratio. So stereochemistry of SN1 reaction is 50% Inversion and 50% Retension. Or, in another word, since they are equal isomeric ratio in the mixture,they are often called as racemic mixture. Hence stereochemistry of SN1 reaction is Racemization.
<u>Reactivity order:</u>
Most suitable substrate molecule for SN1 reaction is tertiary. It is because, it gives carbocation which is most stable, favoring for slow reaction. Hence the reactivity order is
Tertiary>Secondary>Primary

S_N2 Reaction
SN2 stands for **"substitution(S) nucleophilic(N) bimolecular(2) or second order(2)"** reaction**,** which means it will lead to the *displacement* of a group on a molecule, and its rate will depend on the active participation of **two** reactants.
Let us take an example of reaction between Bromomethane and Hydrogen cyanide in presence of suitable solvent molecule, for example: Ethanol or methanol.
$CH_3Br + NaOH \longrightarrow CH_3OH + NaBr$
BromomethaneMethanol
Bromomethane when reacts with sodium hydroxide, methanol as final product is formed.Sodium bromide is formed as a side product.

Mechanism of SN2 Reaction
Bromoethane is a typical primary halogenoalkane. The bromoethane has a polar bond between carbon and bromine. Bromine, therefore, acts as a good leaving group being weak base.

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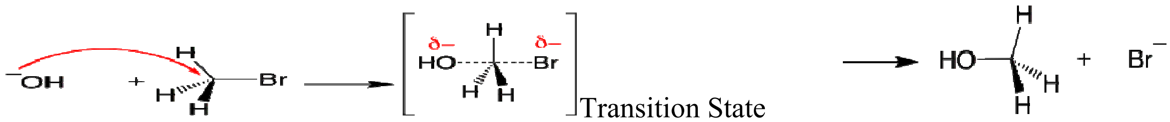
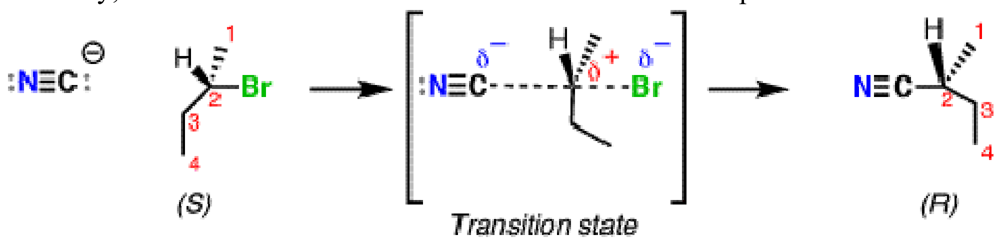
The nucleophile, OH- generated from KOH/NaOHattackes the substrate molecule from the opposite side of the leaving group. Only one side attack is possible since the reaction is fast and nucleophile is attaking the substrate molecule, when there is a leaving group remains in the susbtrate molecule. Therefore, bond breaking and new bond formation takes place simultaneously which gives a highly energetic state called Transition State, is reached. In this state, carbon seems it has 5-bonding state, the C-OH bond is forming and C-Br bond is breaking. So both of them bears small polarity, on OH, -ve charge is disappearing and on Br, -ve charge is forming. As a result, a final stable product, alcool is formed with the liberation of bromide ion.



Nucleophile   Bromomethane   Alcohol, final product

Similarly, reaction between Bromoalkane and CN- is the example of SN2 reaction



Kinetics of SN2 reaction:
The reaction is fast and completes in single step. It means both substrate molecule and nucleophile are involved in this step and both molecules are responsible to control the rate of the reaction. Hence, the reaction is called bimolecular or second order reaction.
Mathematically,
Rate ∞ [Substrate molecule] [Nucleophile]

Stereochemistry of SN2 reaction:
Nucleophile attacks the substrate molecule from back side of the leaving group so only one type of product i.e. inverted product is formed. It means stereochemistry is 100% Inversion.

Reactivity order
Primary>Secondary>Tertiary

Animation and energy diagram for SN1 and SN2 reactions (Youtube)

https://www.youtube.com/watch?v=FhgbWCYa3pQ

Elimination Reaction

Elimination reaction is a type of organic reaction in which two substituent are eliminated from a molecule , in either a one or two step mechanism. The one step mechanism is known as E2 reaction and the two step mechanism is known as E1 reaction. In most organic elimination reactions, at least one hydrogen is lost to form the double bond: in this way, the unsaturation of the molecule increases. If the substrate molecule is single bonded, the final elimination product is double or triple bonded. If the molecule is double bonded, the eliminated product should be triple bonded. This reaction is often known as dehydrohalogination reaction, 1,2 elimination reaction or alfa- beta elimination reaction since the eliminated groups are from 1,2 carbon (alfa, beta carbon). For example:

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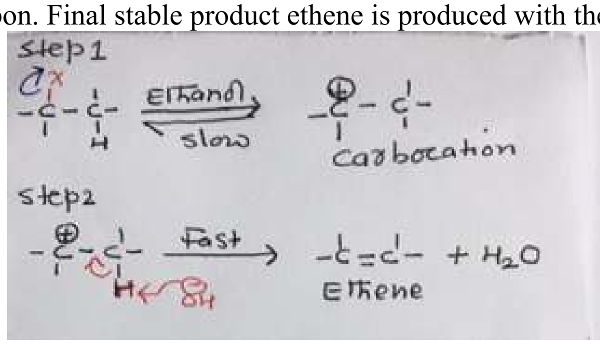
$H_3C - CH_2X + NH_4OH \longrightarrow H_2C{=}CH_2 + H_2O + NH_4X$Haloethane (weak base)
Ethene

Elimination by first order reaction (E1)
This reaction is very similar with SN1 reaction, completes in two steps. The reaction is slow and proceeds without the involvement of base in this step. The base must be weaker in order to slow the reaction.

<u>Mechanism:</u>
Step 1
The most suitable substrate molecule for this reaction is halo ethane. In presence of suitable alcoholic solvent, ionization slowly happens and living group, X- takes away the shared electron pair and forms halide ion. Consequently, intermediate product, carbocation is generated which is highly unstable.

Step 2
The base which is weak for example, NH4OH, in solution gives OH- ion abstract proton from the carbocation in such a way that proton leaves its electron which is utilized for pi- bond formation between carbon and carbon. Final stable product ethene is produced with the formation of water molecule.



Mechanism of E1 reaction

<u>Kinetics:</u>

Rate = k [substrate molecule]

Hence it is unimolecular elimination reaction

Elimination by Second order (E2) reaction:

The second order elimination reaction is very similar with SN2 reaction. This is single step, very fast reaction. Hence, bond breaking and new bond formation takes place very fastly. Therefore, strong base is needed.

$H3C - CH2X + NaOH \longrightarrow H2C{=} CH2 + NaX + H2O$
Haloethane (strong base) Ethene

<u>Mechanism:</u>
The substrate molecule is same as E1 reaction however the nature of base should be strong, for example: NaOH/KOH.The base abstract the proton, proton leaves its electron, formation of pi bond between carbon and carbon and departure of leaving group takes place simultaneously. Hence highly energetic state, called transition state is reached, in which carbon seems it has five bonding stage, which is highly energetic and called as Transition State. Once T.S. reached, leaving group completely departs and final eliminated product, alkene is formed with the formation of water molecule as side product.

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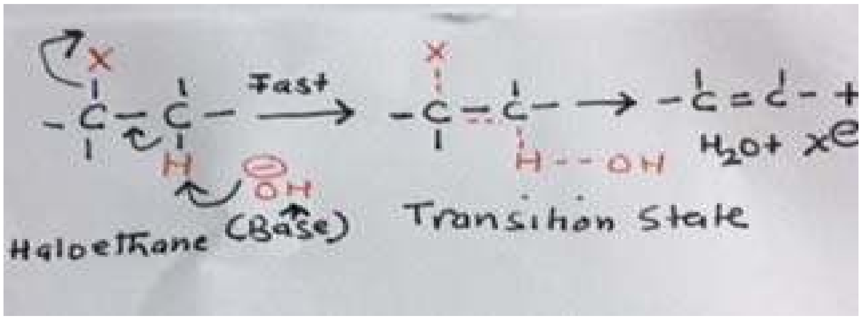
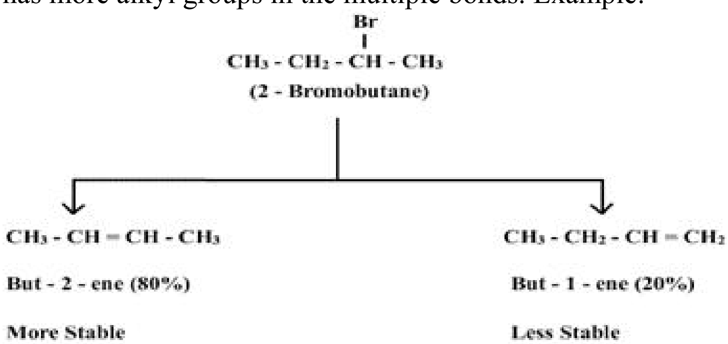


Mechanism of E2 Reaction

<u>Kinetics:</u>
Rate $\propto$ [Substratemolecule][Base]
Hence the reaction is known as bimolecular reaction

Saytzeff Rule:
While there is elimination reaction in the unsymmetrical molecules, the preferred product is one which has more alkyl groups in the multiple bonds. Example:



So, the preferred product is But-2-en which has two alkyl groups where as in But-1-en, only one alkyl group is there.

The SN2 and E2 mechanisms differ in how the R group affects the reaction rate. As the number of R groups on the carbon with the leaving group increases, the rate of the E2 reaction increases

The eliminated product is always more unsaturated in compare to the substrate molecule.

Electrophilic Aromatic Substitution (EAS) Reaction

When substitution reaction occurs in aromatic compounds like benzene molecule by electrophilic reagent, the reaction is known as electrophilic aromatic substitution reaction. For example: Benzene when reacts with nitric acid in presence of sulphuric acid, nitrobenzene as a final product is obtained. This reaction is an example of Nitration reaction, a type of EAS reaction.

Why EAS reaction possible in Benzene molecule?
Benzene molecule has double and single bond in alternate fashion. The pi bonds are weaker and can easily delocalized. Due to which polarity can be generated. This pi electrons give benzene molecule basic nature and the electron deficient group, electrophile therefore favors to attack, which is illustrate below:

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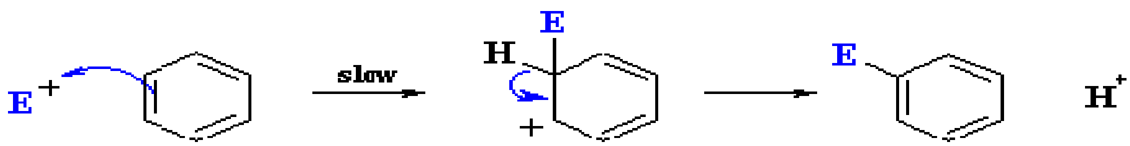


There are 4 types of EAS reactions which are:

1. Nitration reaction
2. Sulphonation reaction
3. Halogenation reaction
4. Friedel -Craft's reaction

1. Nitration reaction

Formation of Nitrobenzene from Benzene is an example of Nitration reaction. Nitro group is inserted into benzene ring by the displacement of H+ ion from the ring.

Mechanism:

It involves three steps:

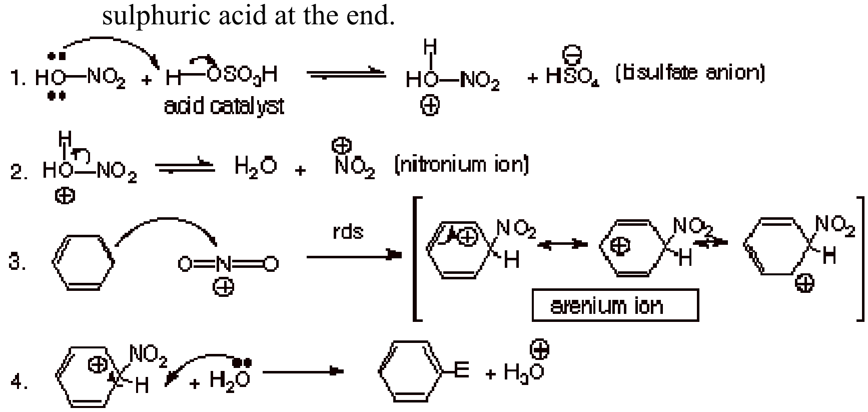
- Reagent generation step: In this step, suitable reagent for this reaction is generated. For this reaction nitroniun ion and base is necessary which should be generated instantly. Therefore, when fuming acids of HNO3 and H2SO4 are mixed together in 1:3 ratios, it slowly ionizes to give nitronium ion, hydronium ion and hydrogen sulphate ion. Nitronium ion is an electrophile for nitration reaction and hydrogen sulphate ion act as a base.

$HNO_3 + H_2SO_4 \longrightarrow NO_2^+ + HSO4^- + H_3O^+$

Fumingacids (1:3 ratio) Nitronium ion

- Aromatic attack to electrophile: The pi-electrons of benzene ring attack electrophile, Nitronium ion so that benzene carbocation is formed which is in resonance form. It is unstable and formed as intermediate product.

- Abstraction of proton by a base: The base HSO4- thus generated, abstract proton from the ring in such a way that proton leaves its electron for the formation of pi -bond between carbon and carbon of the ring. As a result Nitrobenzene is formed with the regeneration of sulphuric acid at the end.



2. Sulphonation Reaction

The end product of sulphonation reaction is Benzene sulphonic acid. Therefore, formation of benzene sulphonic acid from benzene is known as sulphonation reaction. It involves following steps to complete:

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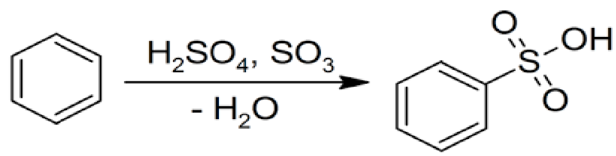


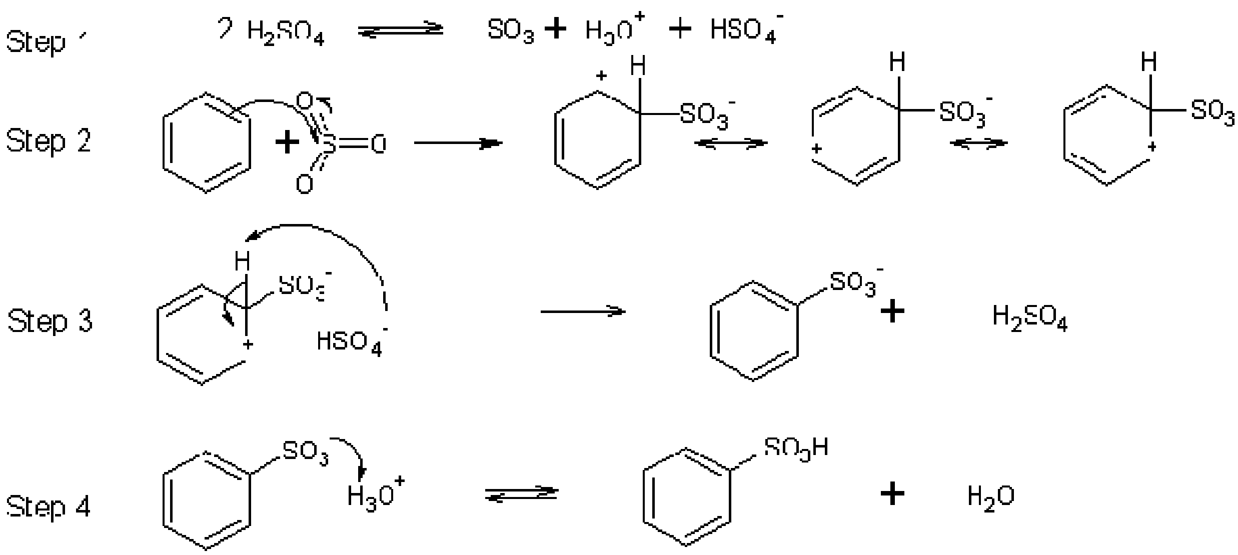
- Reagent generation step: When two molecules of dilute sulphuric acid is treated, sulphur trioxide molecule, hydrogen sulphate ion and hydronium ions are generated. Sulphur trioxide molecule acts as an electrophile for this reaction whereas hydrogen sulphate ion act as a base.

$$2H_2SO_4\,(dil) \longrightarrow SO_3 + HSO^- + H_3O^+$$

- Aromatic attack on electrophile: The pi-electrons attack on electrophile, SO3 molecule so that benzene carbocation is formed, which is slow and intermediate step.

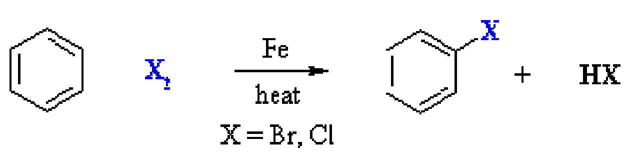
- Abstraction of proton by base: The base H2SO4- abstract proton from the ring in such a way that proton leaves its electron which is used for pi bond formation between carbon and carbon of the ring. A benzene sulphur trioxide ion is formed which is unstable. Since it bears –ve charge, it acts as base.

- Abstraction of proton by base: The benzene sulphur trioxide ion acts as a base which abstract proton from hydronium ion thus generated. So that benzene sulphonic acid is formed at the same time water molecule is generated. In this way final substituted product benzene sulphonic acid is formed. The various steps of the reaction is summarized below:



3. Halogenation Reaction

When hydrogen of benzene ring is displaced by halo group so that halo benzene as final product is formed , the reaction is known as halogenation reaction. Like other reactions, it also involves different steps as follows:



Benzene Halobenzene

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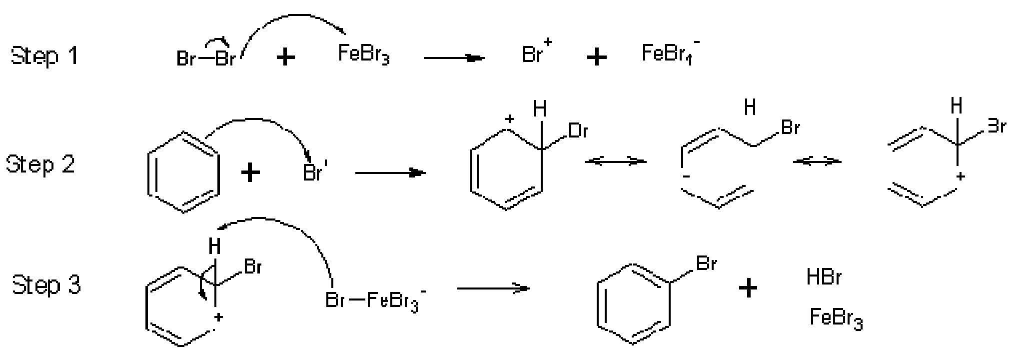
- Reagent generation step: At this step suitable reagents are generated for this reaction. The halonium ion, and base is generated by treating chlorine molecule with iron trichloride molecule

  $Cl_2 + FeCl_3 \longrightarrow Cl^+ + FeCl_4^-$

  Chloroniumion
- Aromatic attack on electrophile: The pi electrons attack electrophile, chloronium ion so that benzene carbocation is formed which is an intermediate product.

- Abstraction of proton by base: The base FeCl4 - abstract proton from the ring so that electrons are used for pi bond formation in the ring as a result resonance of benzene is re-established and final product halobenzebne is formed



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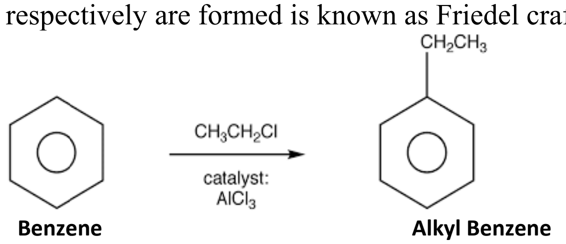
4. Friedel –Craft's Reaction

The name of the reaction is given by the Chemist, Friedel-craft. The insertion of alkyl or acyl group in the benzene ring by the displacement of Hydrogen atom so that Alkyl or Acyl benzene respectively are formed is known as Friedel craft's reaction. This reaction is divided into two types:



i. Alkylation Reaction:

The end product is Alkyl benzene. It involves following steps

- Reagent generation step: When Friedel craft's reagent AlCl3, is treated with alkyl chloride, RCl, Electrophile R+ is generated with the formation of base AlCl4-

  $RCl + AlCl3 \longrightarrow R^+ + AlCl4^-$

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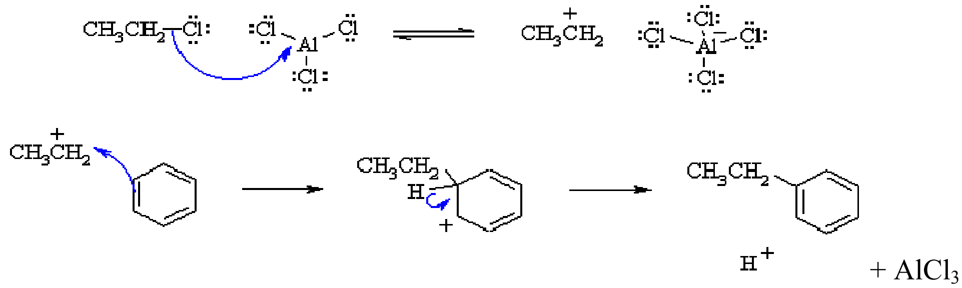
- Aromatic attack on electrophile: The pi electrons of benzene ring attack the electrophile slowly so that benzene carbocation as an intermediate product is formed which is unstable.

- Abstraction of proton by base: The base AlCl4- abstract proton from the ring so that pi bond is formed in the ring and resonance of benzene ring is re-established and the final substituted product alkyl benzene is formed.



ii. **Acylation Reaction:**

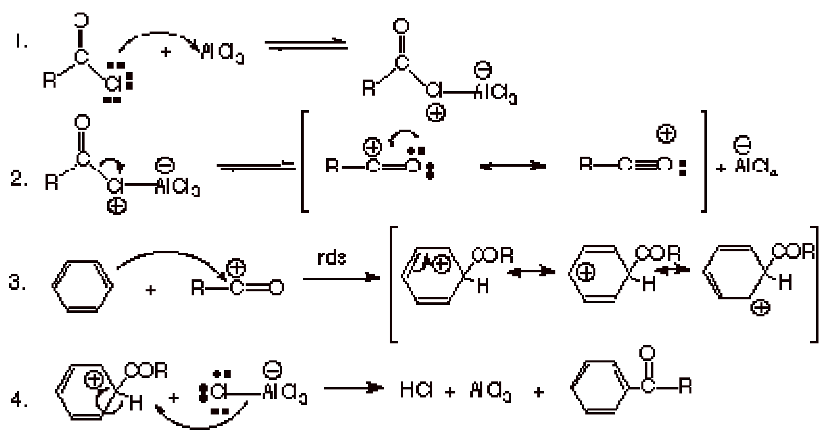
This is same as alkylation reaction only the regent is different. Instead of alkyl benzene , we should take acyl benzene.

- Reagent generation step: When Friedel craft's reagent AlCl3, is treated with acyl chloride, RCOCl, Electrophile R+ is generated with the formation of base AlCl4-

$$RCOCl + AlCl_3 \longrightarrow RCO^+ + AlCl_4^-$$

- Aromatic attack on electrophile: The pi electrons of benzene ring attack the electrophile slowly so that benzene carbocation as an intermediate product is formed which is unstable.

- Abstraction of proton by base: The base AlCl4- abstract proton from the ring so that pi bond is formed in the ring and resonance of benzene ring is re-established and the final substituted product acyl benzene is formed as follows:



Addition Reactions

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When some atom or group of atoms is added in the unsaturated carbon compounds so that more saturated product is formed, the reaction is known as addition reaction. There must be multiple bond (double or triple) in the substrate molecule in order to add acid or any other groups.

There are two types of addition reaction viz: Electrophilic and Free radical addition reactions.

Electrophilic Addition Reaction

If the addition reaction occurs due to the involvement of electrophilic reagent, the reaction is known as Electrophilic addition reaction. For example:

$CH_2 = CH_2 + HX \longrightarrow CH_3\text{-}CH_2X$ [where HX isacid, HCl, HBr, HI, HNO_3.H_2SO_4, HNO_3 etc.]

Therefore, addition of acids to unsaturated carbon compound is the example of electrophilic addition reaction. Acid on ionization generates $H^+ X^-$ ions. H+ ion acts as an electrophile which initiates the addition reaction.

Mechanism:
This reaction completes in two ways, the slow and fast reactions.
$CH_2 = CH_2 + HCl \longrightarrow CH_3 - CH_2Cl$

Slow, two step mechanism:

This is slow reaction and completes in two steps. Let us take the example of addition of Hydrochloric acid on Ethene molecule. The first step is slow ionization step in which. H+ ion generated from acid, is attacked by pi electrons of substrate molecule, Ethene (Actually first, the pi bond of Ethene molecule since it is weaker, sifted to one of the carbon, thus making polar carbons). So that Ethane carbocation as an intermediate product is formed.
Thus generated ethane carbocation is attacked by Cl-(generated from acid)which acts as nucleophile in this step. So that final addition product, Chloroethane is produced, without the loss of anything from substrate molecule.

Step 1
CH2 = CH2 + H^+ ↔ $CH3CH2^+$
Ethen Slow Ethane Carbocation

Step2
$CH_3CH_2+ \ + Cl- \xrightarrow{\text{Fast}} CH_3CH_2Cl$

Single step mechanism:

Sometimes, the reaction also completes in single step which is fast reaction.Since it is fast reaction, bond breaking and new bond formation takes place together and hence highly energetic state called, Transition State is reached. In this state, pi bond between carbon and carbon is not completely ruptured, and new bond formation between carbon and hydrogen and carbon and chlorine is under process. As a result, carbon seems it is pentavalent and therefore, very unstable, which is even difficult to trap. Once transition state is reached, very quickly, stable product Chloroethane is formed.

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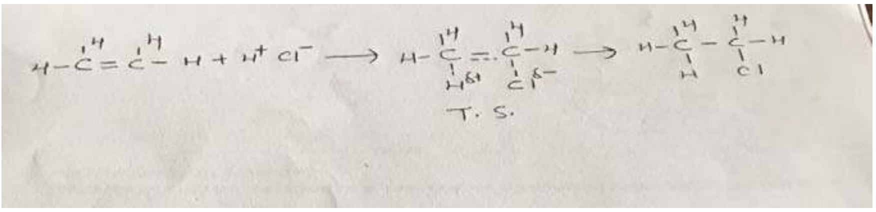


Addition of Acid to Ethene molecule

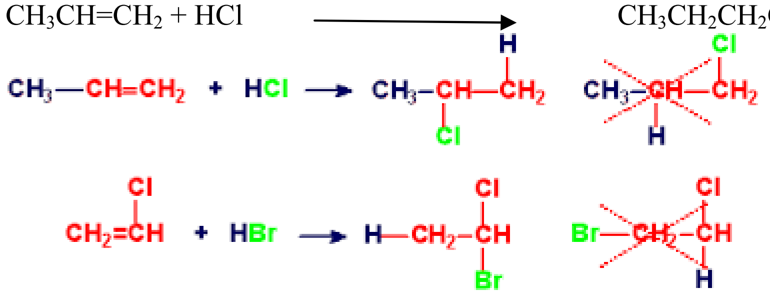
Markovnikoff's Rule:
The addition reaction is possible in any chain system of carbon which are unsaturated. If carbon carbon chain is symmetrical(it means if both the carbon which are multiple bonded, bears identical atoms), for example Ethene molecule having two hydrogen in both double bonded carbons. In this system, the addendum, H+Cl- can attach randomly to the carbons. If H+ goes first carbon, Cl- goes to another carbon, no matter which goes where? But, there is vast number of organic molecules which are not always symmetrical. If there is unsymmetrical unsaturated system, the addition of HCl is not random. It must follow certain rule which is given by a Russian Chemist, Markovnikoff.

Statement of Markovnikoff's Rule:

"*While there is addition of acids to the unsaturated unsymmetrical system, the positive part of acid (H+)goes to the carbon atom which already consist of higher number of hydrogen atom/s*".
In another way, "*While there is addition of acids to the unsaturated unsymmetrical system, the negative part of acid(Cl-)goes to the carbon atom which already consist of lower number of hydrogen atom/s*".
Illustration:

$CH_3CH{=}CH_2 + HCl \longrightarrow CH_3CHCl\,CH_3$ Correct

$CH_3CH{=}CH_2 + HCl \longrightarrow CH_3CH_2CH_2Cl$ Incorrect



Free Radical Addition Reaction

Free radicals are the species having free electrons, hence highly unstable. They are generated by hemolytic bond fission. The best way to generate free radicals is by photolytic reaction (reaction with sunlight). Thermal and radioactive reaction also generate free radicals. When addition reaction is possible by the generation of these free radicals, the reaction is known as free radical addition reaction.

Photolytic reaction
CH4 $\longrightarrow$ CH3° + H° (methyl free radical and hydrogen free radical)
Homolytic bond fission

<u>Mechanism of free radical addition reaction:</u>
For free radical reaction some stimulator is necessary to initiate the reaction by generating free radical. Organic peroxide is taken for this purpose. It generates organic peroxide free radical which initiates the

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reaction. Example of organic peroxide is: Tertiary butyl peroxide, (CH3)3 COO(CH3)3C. Free radical addition reaction involves three steps to complete which are:

1. Chain initiation step
2. Chain propagation step
3. Chain termination step

Let us take Bromination reaction for Methane molecule. The reaction completes in three steps:

1. Chain initiation step
   The organic peroxide molecule by homolytic bond fission, generates organic peroxide free radical. It reacts with chlorine molecule to generate chlorine free radical.
   $$O.P. \xrightarrow{hv} 2R^{\circ}$$

2. Chain propagation step:
   $$H\text{-}Br + R^{\circ} \longrightarrow H\text{-}R + Br^{\circ}$$
   $$H3CCH = CH_2 + Br^{\circ} \longrightarrow H_3CHC^{\circ} - CH_2Br$$
   $$H_3CHC^{\circ} - CH_2Br + H\text{-}Br \longrightarrow H_3CHCH_2CH_2Br + Br^{\circ}$$

Thus generated free radical propagates the reaction by producing new and mew free radicals in each step, hence the name propagation step. The reaction continues until favourable condition. It means if light is sufficient and reagents are enough for the reaction, it continues to produce free radicals.

3. Chain termination step
   Under unfavourable condition, if there are no light or insufficient reagents, the reaction is terminated by randomply colliding one free radical to another forming stable product. Hence reaction stops.
   For example:
   $$CH_3^{\circ} + Br^{\circ} \longrightarrow CH_3Br$$
   $$Br^{\circ} + Br^{\circ} \longrightarrow Br_2$$
   And so on.
   In this way reaction terminates.

Application of Free radical reaction: Explanation of Anti Markovnikoff's Rule
In the presence of organic peroxide, free radical addition reaction favors. It is true when HBr is added to propene in the presence of organic peroxide. Mechanism of this reaction is Free radical addition rather than electrophilc addition reaction. This reaction do not follow Markovnikoff's rule, instead it opposes the rule and hence called as Anti Markovnikoff's rule. It states that:
"*While there is addition of Hydrobromic acid to the unsaturated unsymmetrical system in the presence of organic peroxide, the positive part of acid(H+) goes to the carbon atom which has least number of hydrogen atom and vice versa"*
If there is absence of organic peroxide, it goes with Markovnikoff'r rule. Note that Anti Markovnikoff's rule is only applicable to HBr in presence of OP. Whatever the condition, all other acid added following Markovnikoff's rule.
Addition of HBr to propene molecule in presence of oP follows free radical addition reaction while addition of other acids follows electrophilic addition reaction.

<u>Mechanism:</u>

Step 1: Chain initiation step:

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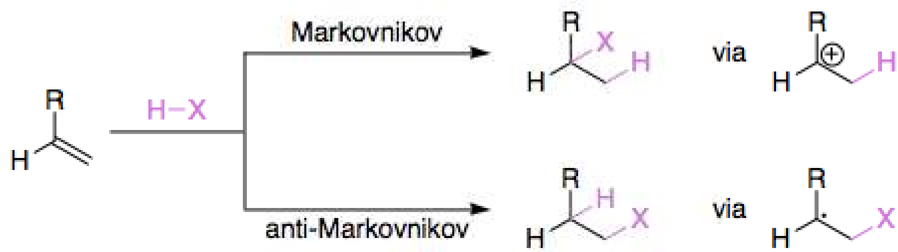
Organic peroxide (OP) $\longrightarrow$ 2R°

Step2: Chain propagation step:

R° + H-Br $\longrightarrow$ H-R + Br°
$CH_3CH = CH_2$ + Br° $\longrightarrow$ $CH_3C°HCH_2Br$
$CH_3C°HCH_2Br$ + H-Br $\longrightarrow$ $CH_3CH_2CH_2Br$ + H°

Above reaction continues until favourable condition
Step2: Chain termination step:

During unfavourable condition, reaction terminates when two generated free radicals reacts with each other randomly forming stable products and thus reaction terminates.
For example:
H° + Br° $\longrightarrow$ HBr
Br° +Br° $\longrightarrow$ Br_2
Thus reaction terminates.
[Note: Addition of HBr to Propene in presence of organic peroxide follows Anti Markovnikoff'sRule because it generates 2° carbon free radical which is more stable in compare to 1°carbon free radical if added following Markovnikoff's rule.]



Questions:

1. Complete the reactions and write their mechanisms:
   - **i.** CH_3Br + NaOH $\longrightarrow$
   - **ii.** $CH_3C_2H_5C_3H_7CBr$ + NaOH $\longrightarrow$
   - **iii.** C_6H_6 + H_2SO_4 + HNO_3 $\longrightarrow$
   - **iv.** C_6H_6 + H_2SO_4 $\longrightarrow$
   - **v.** C_6H_6 Cl_2+ $FeCl_3$ $\longrightarrow$
   - **vi.** C_6H_6 + RCl+$AlCl_3$ $\longrightarrow$
   - **vii.** C_6H_6 +RCOCl+ $AlCl_3$ $\longrightarrow$
   - **viii.** $H_2C=CH_2$ + HBr $\longrightarrow$
   - **ix.** $H_3CCH = CH_2$ + HBr + Organic peroxide $\longrightarrow$
   - **x.** $H_3XC – CH_3$ + NH_4OH(Base $\longrightarrow$
   - **xi.** $H_3XC – CH_3$ +NaOH(Base) $\longrightarrow$

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UNIT 6 INDUSTRIAL CHEMISTRY

Written by**: Prof. Sabitri K. Tripathi**

Objective:
To make understand the chemistry of different engineering materials, their properties and manufacture procedures

Applications :

- **Construction**
- **Decoration and protection materials**
- **Military uses**

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1. CEMENT

Cement is the engineering construction material which binds aggregate, sand and bricks together. The raw materials used for preparing cements are available in the earth's crust. They are mainly lime stone and clay. Cement has very good adhesive property when water is added in it, therefore it is called as hydraulic cement.

Classification of cement:

Natural cement: Limestone and clay forms the major constituents of cement which are found naturally. Under pressure and temperature, these materials convert into hard calcined mass which when combines with silica and alumina to form calcium and aluminium silicates. This is known as natural cement. It has property similar to OPC cement but weaker and easily hydrated.

Puzzolona cement: This is the oldest man made cement, used during roman periods for making concrete walls. Puzzola means volcanic ash, when mixed with silica alumina and calcium oxede, puzzolona cement is formed. This is also hygroscopic and weaker in nature.

Portland cement: This is synthesized cement.The invention of Portland cement is attributed to Joseph Aspdin, England in 1824. He baked clay and limestone mixture and grinded into fine powder. Upon addition of water into it, it acts as cement. He called it Portland cement, because its property resemples with the stone found in Portland.

Composition of Portland cement**:**

The major constituent of Portland cement is Lime stone which is known as Calcarious mass and Clay, the Argillacious mass. Chemical composition of clay is mixture of alumina, silica and iron oxide.The chemical constituents of Portland cement is as follows:

Compounds/	%
Lime(CaO)	60-67
Silica(SiO_2)	17-25
Alumina(Al_2O_3)	3-8
Iron oxide	0.5-6
Magnesia(MgO)	0.1-4
Sulphur trioxide(SO_3)	1-3
Alkali oxide	0.5-1.3

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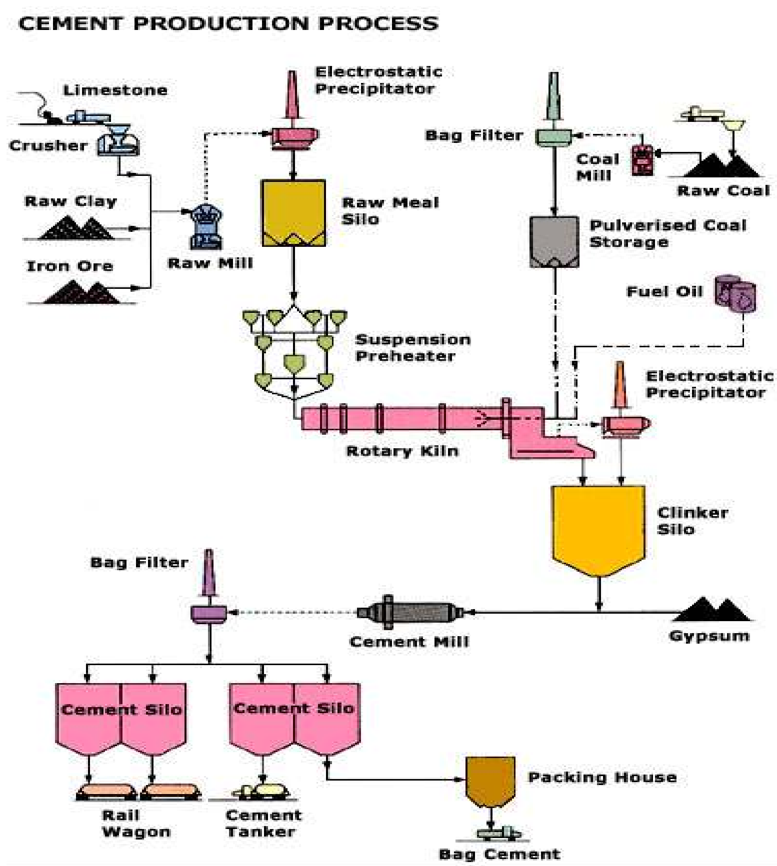
Table: Composition of Portland cement

Compound	Formula	Short form
Calcium oxide	CaO	C
Silicon oxide, Silica	SiO2	S
Aluminium oxide, Alumina	Al2O3	A
Iron oxide	Fe2O3	F
Water	H2O	H
Sulphite	SO3	S

Table: Short form of chemicals used in Portland Cement manufacture

Manufacture of OPC Cement



OPC cement can be manufactured by two ways:

1. Dry Process
2. Wet Process

There are mainly 5 steps involved in the manufacture of OPC cement for both processes,

1. <u>Crushing and grinding of raw material</u>

First the raw materials are crushed and grinded into small suitable size particles.For dry process, the raw materials are dried up before crushing.

2.<u>Mixing or Blending</u>

The raw, grinded material is mixed with clay in the desired proportion and mixed well to get homogenous mixture with the help of compressed air.If it is wet process, water is added to make slurry, having 35-40% water.

3. <u>Heating</u>

The slurry or the dried mass is passed into a heating chamber in the kiln where there is different temperature maintained zones:

<u>4. . Drying zone:</u> In this zone, 400 oC temperature is maintained. When slurry /or dried raw materials comes in contact in this zone, all the water molecules gets evaporated and makes it completely dry and clay is broken down into alumina, silica and iron oxide.

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5. Calcination zone: At 800-1000 oC, oxidation of limestone occurs in the absence of oxygen which is known as calcination so that quick line is produced.

$CaCO_3 \longrightarrow CaO + CO_2$

4. Clinkeringzone: At 1500-1800 ^oC, Alumina, silica and iron oxide gets fused with quick lime forming granular mass of 0.5-1cm diameter called clinkers. There is cooling mechanism to cool the clinkers which is collected for further process.

$2CaO + SiO_2 \longrightarrow Ca_2SiO_4$ (Dicalcium silicate, C_2S)

$3Cao + SiO_2 \longrightarrow Ca_3SiO_2$ (Tricalcium silicate, C_3S)

$3CaO + Al2O_3 \longrightarrow Ca_3Al_2O_6$ (Tricalcium alumina silicate, C_3A)

$4CaO + Al2O_3 + Fe_2O_3 \longrightarrow Ca_4Al_2Fe_2O_{10}$ (Tetracalciumaluminoferrite, C_4AF)

So, clinker is the mixture of C_2S, C_3S, C_3A and C_4AF.

6. Grinding: Thus collected clinkers are further cooled and 1-3% gypsum salt is added and grind into very fine powder. The process involved in hydration of cement is that, when the water is added into cement, it starts reacting with the C3A and hardens. The time taken in this process is very less, which doesn't allow time for transporting, mixing and placing.
The gypsum reacts with C3A in the cement to form ettringite, Tricalciumsulpho aluminate (Michaelis salt) and this prevents quick setting. The other compounds in the cement i.e. C3S, C2S and C4AF require water for the hydration process to proceed to form the hardened cement stone. If there is no water added, no hydration will occur and there will be no strength development of the cement.Therefore, gypsum salt elongate the initial setting time. It retards the dissolution of C3A by forming tricalciumsulphoaluminate ($3CaO.Al_2O_3.xCaSO_4.7H_2O$) which is insoluble.

$3CaO.Al_2O_3 + xCaSO_4.\&H_2O \longrightarrow 3CaO.Al_2O_3.xCaSO_4.7H_2O$
Gypsum — Tricalciumsulphoaluminate(insoluble)

Thus tricalciumsulpho aluminate prevents the too early setting of cement.

7. Packing: Grinded of powdered mass is packed into airtight sacks, ready for marketing.

Setting and Hardening Mechanism of Cement

Cement only functions when water is added into it. Hydration reaction occurs after addition of water When water is added, exothermic reaction starts with the evolution of heat energy. Gel like mass is formed and some crystals are also seen. This crystal like mass starts to bind the aggregates forming compact masses. Formation of gel gives stiffening effects whereas formation of crystal gives hardening effect of cement. So hardening hardening is the development of strength to the cement. More the crystals greater the strength. The functioning of cement completes in two steps:
Step1:
Initial Setting of cement:
When water is added, hydration reaction of cement occurs and crystal of C3A and gel of C3AF are formed.

$3CaO.Al_2O_3 + 6H_2O \longrightarrow 3CaO.Al_2O_3.6H_2O + 880kj/kg$
Tricalcium aluminate — Hydrated tricalcium aluminate(Crystalline)

$4CaO.Al2O_3.Fe_2O_3 + 7H2O \longrightarrow 3CaO.Al2O_3.6H_2O + CaO.Fe_2O_3.H_2O + 420kj/kg$

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Tricalciumaluminoferrite (Crystalline) (Gel)

Small amount of tobermonite gel is also formed at this state by hydrolysingdicalcium silicate which has very good adhesive property.

$2[2CaO.SiO_2]+4H_2O \longrightarrow 3CaO.2SiO_2.6H_2O + Ca(OH)_2 + 250kj/kg$

Dicalciumsilicate Tobermonite gel Crystallinecalciumhydroxide

Final Setting of cement:

Final setting and hardening of cement means there is the complete formation of hard substance giving full strength to the cement. The gel formed in the initial setting converts into hard substance and crystallisation of calcium hydroxide takes place as end products of hydration of cement.

$2[2CaO.SiO_2]+4H2O \longrightarrow 3CaO.2SiO_2.3H_2O + 3Ca(OH)_2 + 500kj/kg$

Tricalcium silicate Tobermonite gel Calcium hydroxide crystals

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2.PAINTS

***Paint** is a chemical* substanceessentially a coating or covering material applied on metallic or non-metallic surfaces for *decorative* and/or *protective* purposes. These are liquid substance giving different colors when applied to the surface, as a thin layer that forms a solid dry adherent film after oxidation/evaporation/ polymerisation". Good paints should be non toxic to human health and environment friendly. It should be easily available, easy to apply and homogenous. It should be non reactive with air, water and heat and hence long lasting.

Chemistry of Paint

1. **Pigment:** This is the principle constituent of paint which is solid substance. It gives desired colour to the paint. The chemical substance used as pigment are:
   Red color: Ferric oxide, Red Lead, Chrome red etc
   Blue color: Proshian blue
   Green color: Chromium oxide
   Brown color: Brown Umbre
   Black Color: Carbon black
   While color: White lead, zinc oxide, lithophone, titanium oxide

   Characteristics of good pigments: Good pigments should be
   Environment friendly and causes no health illness, it should be cost effective, easy to mix, easy to apply, opaque, and cheap.
2. **Vehicle or drying oil**: This is a chemical substance which suspends or carry pigments and other constituents of paint. These are glyceryl esters of high molecular weight fatty acids derived from plants and animals.
3. **Thinner:** Thinners are used to reduce the viscosity of paint. Hence it helps:
   - Quicken the drying process of paint
   - easy to apply
   - Lower the cost
   - Dissolves the constituents of paint

   Common thinners used are: spirit, naphthalene, kerosene, turpentine, benzene, xylol etc.
4. **Drier:** These are oxygen carrier catalyst. It accelerates the drying process. The chemicals used as drier are resonates, linoleates, tungtates, and naphthenates of Co,Mn,Pb and Zn. These chemicals absorb atmospheric oxygen, hence oxidises, through condensation polymerises, looses water to spped up the drying process.
5. **Extenders or fillers:** These are low refractive index substances added into the paint which are cheaply available so that it increases the volume. In this way, it lowers the cost of paint. Talc, graphite, china clay, limestone, asbestos, slates are examples of extenders.
6. **Plasticisers:** These are chemicals substances added in the paint to increase the elasticity of paint, so that it gives smooth aesthetic to the paint, minimise cracking and gives longer life to the paint. Common chemicals used are:Tricrecyl phosphate, triphenyl phosphate, Tributyl phthalate, diamyl phthalate etc.

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3. EXPLOSIVES

Explosives are the chemical substances which upon application of pressure or, temperature and small shocks, produces tremendous amount of heat, pressure, sound and heat energy exothermically which happens suddenly. Due to rapid reaction within a small confined speace, a high pressure is developed which shatter the rocks, walls etc. If the process is slowe, it can be used to propel rockets and projectiles.
Explosives are chemical substances and they may be mixtures of different chemical substances. In order to explode, the molecule must have low energy of dissociation.It means at least one bond should easily break. It is possible when two combining atom have low or zero electronegativity differences. For example: N-N, N-O, O-Cl, N-Cl bonds.

Generally, explosives have four basic characters:

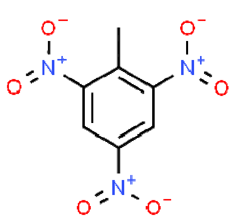
1. It is a chemical compound or mixture ignited by heat, shock, impact, friction, or a combination of these conditions;
2. It decomposes rapidly in a detonation when ignited
3. There is a rapid release of heat and large quantities of high-pressure gases that expand rapidly with sufficient force to overcome confining forces
4. The energy released by the detonation of explosives producesshattering of rocks, walls, ground vibration, air blast.

Classification of Explosives:
Broadly, chemical explosives are divided into two classes:

1. Low explosives: The low explosives explodes through combustion propagating through heat transfer from hotter to colder layer so that it ignites. These are primarily used for propelling. The are the mixtures of readily combustible substances, for example gunpowder, when ignited, undergo rapid combustion.It can be easily controlled since the explosion is not vigprous.
2. High or detonating explosives: These are upon explosion produces large volume, high sound, vibration and releases tremendous amount of gases. Actually high explosives propagates supersonically through shock wavesand thus decomposing a substance extremely quickly.Theyare used mainly for shattering. They are unstable molecules that can undergo explosive decomposition without any external source of oxygen and in which the chemical reaction produces rapid shock waves.Due to this character, they are used for road construction and other construction works as a shattering tool. Important explosives include TNT,dynamite,picric acid. Cyclonite (RDX) was an important explosive in World War II.

Preparation properties and uses of some explosives:

2,4,6 Tri NitroToulene(TNT)



TNT is a yellow crystalline odourless solid, a high explosive that does not occur naturally in the environment. It is made by combining toluene with a mixture of nitric and sulfuric acids. It is one of the

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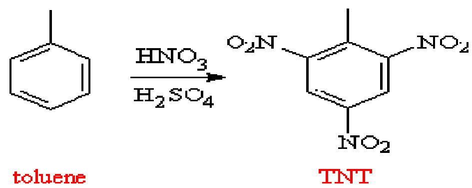
most widely used military high explosives, partly because of its insensitivity to shock and friction. It has been used extensively in the manufacture of explosives since the beginning of the 20th century and is used in military cartridge casings, bombs and grenades . So it is regarded as safest explosive. Currently,it is the widest manufactured explosive in the world.

Manufacture:

This can be manufactured by nitration of toluene. When toulelene is treated with 1:1 ratio of fuming nitric and sulphuric acids with continuous stirring. After some time crystalline pale yellow solids appear which is TNT. Thus produced TNT is not pure. It might consist of unreacred acids. So further purification process is needed.



Thus produced TNT is filtered, washed with distilled water. Then washed with ammoniacal solution of Na_2SO_3. After that wash with distilled water. TNT thus appeared is pure. It is then air dried and poured into a container.

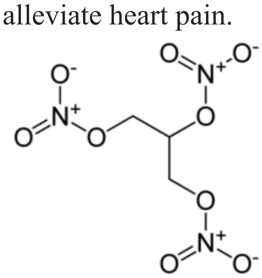
Properties:

Physical properties:
State: Yellow crystalline solid
Odour: Odourless
M.P.: 80.9 ^oC (low melting point
B.P. : 240 ^oC
Chemical properties:
It is Non hygroscopic. It is inactive with most of the metals. It cannot react with air. Due to these properties, it can easy to transport from one place to another. Hence it is known as safest explosive and widely used.
Uses:

1. Road construction
2. Demolishing of old buildings
3. Under water construction
4. Mine industry

Tri NitroGlycerine(TNG)

Nitroglycerin is an organic nitrate which is gel- liquid substance. In its pure form, it is colorless. Nitroglycerin is a nitroglycerol that is glycerol in which the hydrogen atoms of all three hydroxy groups are replaced by nitro groups. It acts as a prodrug, releasing nitric oxide to open blood vessels and so alleviate heart pain.



Tri Nitro Glycerol

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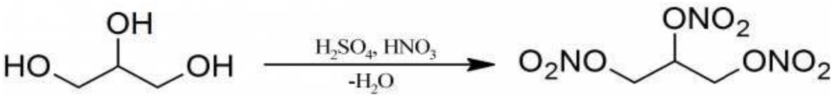
Manufacture:

It was first made by AscanioSobrero in 1846 by treating glycerol with a mixture of nitric and sulphuric acid.



Glycerol Nitroglycerol(TNG)

When glycerol is added to conc. H2SO4(60%) and HNO3(40%) at 10 ^oC with continuous stirring. This nitrating mixture is then poured inot a separating funnel for ppurification purpose. NG is oily gel mass which is heavier than other impurities present. So it goes down in the separating funnel. Its layer can be separated by opening the stopper at the bottom. Thus collected NG washed with distilled water and then with dilute sodium carbonate solution as a dehydrating agent. So that thus obtained NG is pure and stored for future use.

Properties:

Physical properties:
Color: Colorless
Odour: Odourless
State: Oily liquid

Chemical properties: It is highly reactive liquid. It can reac with air, water , metals. Therefore, it is very difficult to transport from one place to another. It is generally kept in saw dust due to its high reactivity with other substances.

Uses:

Road construction and demoshing old buildings

4. LUBRICANTS

Lubricants are chemical substances used between two colliding(rotating) objects to reduce the frictional force to increase the efficiency of the machine so that wear and tear effect is minimized to increase the life span of the object.

Classification:

Lubricants can be classified as:

1. Solid: For example, powder, graphite etc.
2. Semi solid: For example, wax, ghee etc.
3. Liquid: For example, oils, emulsions (soap water, oil water)etc.

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QUESTIONS

1. What is the chemical composition of Portland cement?
2. What are the argilacious and calcarious mass in the cement?
3. What is the function of gypsum salt in the cement?
4. Explain the setting and hardening mechanism of cement?
5. What do you mean by hydration reaction of cement? Why it is omportant?
6. What are the end products of cement after its application?
7. Discuss the different steps of manufacture of OPC cement.
8. What are the chemical constituents of paints?
9. What are the characteristic feature of good paints?
10. Why thinner and driers are added in a paint?
11. What is the function of Filler in the paint?
12. What do you mean by explosives, why it explode?
13. What are the characteristics of explosives?
14. What is the basis for classification of explosives?
15. Write the preparation properties and uses of TNT and TNG.
16. Define and classify lubricants giving at least two examples each.

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UNIT 7 ENVIRONMENTAL CHEMISTRY

Written by**: Prof. Sabitri K Tripathi**

Objective:
To enrich the knowledge of environment and its protection so that students can apply the gained knowledge to work with environment friendly innovation and practices and to think the ways to save the planet earth.

Application:
Environmental Chemistry deals with the study of environment and the life in the earth. The knowledge will be used by various agencies, researchers and professionals to explore nature and the sources of pollutants and their control measures and to use tools and techniques which are environmental friendly.

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ENVIRONMENT

It is the nature due to which life is possible on earth.The biotic and abiotic elements which surrounds and interact with us is therefore, envirobment which comprises the air, water, forest, plants. Soil and rocks. The nature is the creater and driver of the environment and therefore, called as natural environment for all living organisms. However, due to various anthropogenic activities, it has been deterioting and it is the biggest thread to the environmentalist to conserve it. Environment comprises following four segments:

1. Lithosphere
2. Hydrosphere
3. Atmosphere
4. Biosphere

1. Lithosphere

The word Lithosphere is derived from greek word, litho meaning stone which means the hard substances on earth's crust for example, soil, stones, rocks minerals are the elements of lithosphere which forms earth's crust. The lithosphere mainly contains three layers –
Inner and Outer Core**:** Central fluid or vaporized sphere of diameter of about 2500km from the centre.
Mantle:It is about 2900-3000 km above the core in molten state.
Crust:Outermost solid zone about 8-40 km above mantle.

***2.* Hydrosphere**

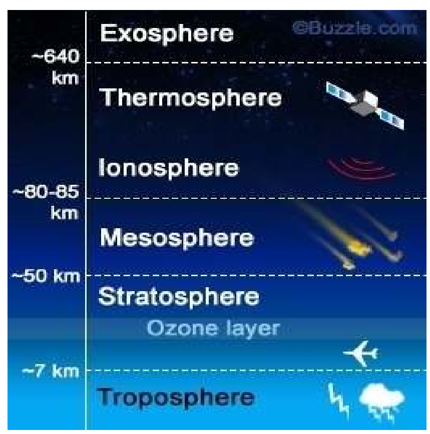
It covers more than 75% of the earth surface either as oceans or as fresh water. Hydrosphere includes sea, rivers, oceans, lakes, ponds, streams etc.97% of total water is saline, 2% comprises snow, ice caps and remaining 1% comprises river, lakes and stream waters which is used for drinking and domestic purposes.

3. Biosphere

This segment of environment consists of atmosphere (air- 0_2, N_2, CO_2). Lithosphere (land- minerals, salts, food, nutrients) and hydrosphere (water- dissolved oxygen, salts) which influences and support the entire biotic and abiotic life systems.

4. Atmosphere

The air surrounding the earth is known as Atmosphere. It extends to the height of about 1600 km from the earth surface. It consists of oxygen, carbon dioxide, nitrogen and other inert gases. There are different layers of atmosphere;



<u>Troposphere</u>: This layer is closest from the earth's crust and hence is the lowest layer which comprises all the gases. It determines the temperature, cloud, rain, snow. The troposphere contains about 75% of all of the air in the atmosphere, and almost all of the water vapour (which forms clouds and rain). The decrease in temperature with height is a result of the decreasing pressure. This is the reason that air higher up is cooler than air lower down.

<u>Stratosphere: </u>This extends upwards from the troposphere to about 50 km. It contains much of the ozone in the atmosphere. The increase in temperature with height occurs because of absorption of ultraviolet (UV) radiation from the sun by this ozone. Temperatures in the stratosphere are highest over the summer pole, and lowest over the winter pole.By absorbing dangerous UV radiation, the ozone in the stratosphere protects us from skin cancer and other health damage.

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Mesosphere: The region above the stratosphere is called the mesosphere. Here the temperature again decreases with height, reaching a minimum of about -90°C at the "mesopause The thermosphere lies above the mesopause, and is a region in which temperatures again increase with height. This temperature increase is caused by the absorption of energetic ultraviolet and X-Ray radiation from the sun.

Thermosphere and ionosphere: The region of the atmosphere above about 80 km is also caused the "ionosphere". Because of high solar radiation, molecules are converted into ions with positive charge. The temperature of the thermosphere varies between night and day and between the seasons. The Ionosphere reflects and absorbs radio waves, allowing us to receive shortwave radio broadcasts.

Environmental Pollution

Environment is created by nature. The air, water, soil, flora and fauna are elements of environment. The planet earth is the home for all living organisms and human beings are the most intelligent organism here. Their population is7.7 billion, crossing the limit to hold them on this planet earth. For this large population, human beings are continuously developing new and new technologies to make their life easier. They are going blindly for massive production and unhealthy competition locally and globally, without caring nature. In the name of development, deforestations, fossil fuel burning and mismanagement of natural resources is everywhere. Use of harmful chemicals in the farm, hospitals, industries, food production etc. is common. These are very unhealthy practices made by human beings in the name of development. Air, water and soil has been contaminated, altering with their natural composition and thus harm to the living organisms, is known as **Polluted Environment.** Human beings are the major cause of it. So by different anthropogenic activities, environmental pollution is vigorously observing and it is the great concern to environmentalists and all human beings.

Natural Pollution: Natural calamities are one of the causes of environmental pollution. For example, volcanoes eruption produces lot of poisonous gases, the lava –river further degrade the environment. Similarly, floods, hurricane, cyclone are the natural calamities destroying life and environment.

Man made: Major contributors are the activities of human beings for the environmental pollution. The increased population and unmanaged natural resources are the causes of man induced pollution. Consequently, air has been poisonous to breathe, water quality is continuously degrading and scarce and soil fertility is degrading.

Air Pollution

Simply the poisonous air to breathe is known as polluted air. Alteration of air quality with its natural composition, due to which it causes negative effect to the living beings, is called polluted air. Therefore, air pollution means the undesirable presence of impurities or the abnormal rise in the proportion of some constituents of the atmosphere.

The healthy air consists of:

Nitrogen	78.1%
Oxygen	20.9%
Argon	0.93%
Carbon dioxide	0.03%
Neon	0.0018%

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Helium	0.00052%
Methane	0.0002%
Krypton	0.00011%
Hydrogen	0.00005%
Xenon	0.00001%

Effect of Air Pollution
Effects of air pollution are local, regional as well as global level. Acid rain is localized where as ozone layer depletion may be regional and global. Similarly green house effect is global effect of air pollution.

OZONE LAYER DEPLETION

Ozone (O_3)is found in the atmosphere from 10-40km from the earth's surface in the stratosphere and troposphere region, mainly in the stratosphere region which comprises 90% of total ozone. It is a protective layer for harmful radiation entering into earth's crust. It helps by not allowing harmful radiations for example UV rays. The UV rays are dangerous when reaches to earth's crust and get trapped which is the cause of many diseases to human.
Ozone is unstable compound of oxygen. It can easily breaks into oxygen molecule (O2) and oxygen atom(O). When plastics and other coolants produces CFCs, which when escapes into the environment, causes to destroy ozone molecules. It is as much dangerous that one molecule CFC can destroy I lakh molecules of ozone. Therefore, CFC is the main cause of ozone layer destruction. In some places of earth, for example, polar region, due to accumulation of CFCs, the layer becoming more and more thinner, it means depleting the layer. Ozone "hole" is a reduction in concentrations of ozone high above the earth in the stratosphere. The ozone hole is defined geographically as the area wherein the total ozone amount is less than 220 Dobson Units.
Therefore, man-made chlorines, primarily chloroflourobcarbons (CFCs), contribute to the thinning of the ozone layer and allow larger quantities of harmful ultraviolet rays to reach the earth. Human activities is one of the major causes of ozone depletion. The release of various compounds containing chlorine or bromine, accounts for approximately 75 to 85 percent of ozone damage which are mainly used in refrigerators and coolants.

Mechanism of Ozone layer depletion:
When CFCs are released into the atmosphere, it reaches to the stratosphere region of the atmosphere where it gets in contact with UV rays and reaction starts as follows:

$$CFCs + O_3 \xrightarrow{hv} ClO + O_2$$

$$O + ClO \xrightarrow{hv} Cl + O2$$

Thus produced chlorine atom helps to destroy ozone vigorously since the reaction is continuous.
Other harmful compounds which destroy ozone layer are bromine compounds, some nitrogen compounds like nitrous and nitric oxide.

Effects:

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Ozone acts as a protective shield to protect the Earth's surface by absorbing harmful ultraviolet radiation. If this ozone becomes depleted, then more UV rays will reach the earth. Exposure to higher amounts of UV radiation leads to serious health impacts to human beings, animals and plants, which are as follows:

(a) Harm to human health: Skin cancers, sun burns. cataracts, blindness and other eye diseases and cancers

(b) Adverse impacts on agriculture, forestry and natural ecosystems: Major crop species are vulnerable to increased UV, resulting in reduced growth

(c) Animals: In domestic animals, UV over exposure may cause eye and skin cancers.

It is believed that when stratosphere becomes colder, there is high risk of ozone depletion. Therefore, polar regions are high risk for ozone hole. Increased global warming traps heat in the troposphere, so less heat reaches the stratosphere which makes it cold and favorable for ozone depletion. Therefore, greenhouse gases act as a blanket for the troposphere and making stratosphere colder.

Control Measures

- Strict rules and regulation for not using plastics
- Introduction of alternatives for plastics
- Encourage for the use of Bio degradable plastics
- Refrigerators, coolants and spray should be environment friendly
- Raising awareness
- Burning of plastics should be banned

Links of Youtube for Ozone layer depletion

https://www.youtube.com/watch?v=aU6pxSNDPhs

GREEN HOUSE EFECT/GLOBAL WARMING

The phenomenon of heating of earth's surface by different gases, for example carbon dioxide, methane, water vapor is known as global warming or green house effect. The effect is very similar with glass houses in botanical garden. UV rays from sunlight enters into the glass house, trapped by soils, plants and trapped in the glass house, cannot pass away and the room gets heated up and making favorable for plant growth. Similar effects can be seen in earth's crust. The sun rays after striking earth's crust, absorbed by soils, plants and returned back. It is estimated that about 30% rays returned back to the atmosphere and rest absorbed and trapped. Thus helps to warm the earth's surface. This phenomenon of heating of earth's surface is known as global warming or greenhouse effect. The gases contributing for these effects are CO_2, CH_4, and water vapor. Due to this effect, earth is suitable for living otherwise it might be as cool as moon.

Effects of global warming

Global warming is very important for making earth's surface warm, however if global warming contributing gasses concentration in the atmosphere is increasing, the temperature on the earth is increasing and that is not suitable for living organisms. Today, human are causing to increase the green house gases concentration and rising the temperature of the planet rapidly. It is recorded globally that 2014-2018 is the hottest years globally. If the trend continue at the current rate, global warming will reach 2.7 degree Fahrenheit(1.5 degree Celsius) above preindustrial levels between 2030-2052.

The increased global warming causes to change the climate system in many ways:

- Causes extreme weather conditions for example: floods, cyclones, heat waves, droughts hurricanes etc.
- Unusual precipitation and droughts
- Melting of ice and glaciers

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- Sea level rise and temperature of sea increases
- Affecting to the whole ecosystem. Internal migration, diseases outbreaks affecting water systems, land system and productivity.

In this way whole ecosystem is affected and life is at risk. Due to temperature rise, outbreaks of different diseases like dengue, jicaetc occurs. Floods and sea level rise causes further killing life. Drought and too much rain causes less production of food. In this way there is food security problem arises and life is tougher to survive.

Control Measures:
Man is the sole cause of increased global warming. They should stop doing any activities which are against the nature. Therefore, following should be taken care of:

- Plant is only organism which absorbs the increased carbon dioxide caused due to human activities. Therefore, stop cutting trees. Encourage to grow more and more trees.
- Reduce the consumption of fossil fuels such as coal and petroleum. Encourage to use alternative source of fuel, for example: solar energy, hydro-energy, bio energy etc.
- Reduce emission of CFCs by introducing environment friendly alternatives
- Go with green chemistry

Youtube for Global warming
https://www.youtube.com/watch?v=oJAbATJCugs

ACID RAIN

When the atmosphere is healthy, rain drops on the earth's surface which is slightly acidic due to the presence of CO2 in the atmosphere, making carbonic acid. But when it is polluted, different increased concentration of gases there reacts with water forming acids which falls on the earth's crust in the form of rain, which is known as acidrain.
The polluted air consists of higher concentrations of CO_2, SO_2 and NOx which combines with water forming their respective acids as follows:

- The oxides of sulphur react with water to form H_2SO_4

$SO_2+O_2 \longrightarrow 2SO_3$

Thus formed SO2 reacts with water in air forming H_2SO_4

$SO_3+H_2O \longrightarrow H_2SO_4$

- The oxides of nitrogen undergoes photochemical and chemical reactions forming nitric acid

$NO+O_3 \longrightarrow NO_2+O_2$
$NO_2+O_3 \longrightarrow NO_3+O_2$
$NO_2+NO_3 \longrightarrow N_2O_5$
$N_2O_5+H_2O \longrightarrow 2HNO_3$

- Carbon dioxide reacts with water forming carbonic acid

$CO_2+H_2O \longrightarrow H_2CO_3$

In this way various types of acids are formed due to increased level of air pollution.

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Effects:
Agriculture: It damages plants and soil character. When acid rain falls into the soil, it turns soil acidic. Most of the plant requires slightly basic soil. Therefore, fertility capacity of the soil is destroyed and needs soil treatment action. Further, when it falls on the plants, the leaves and stem are damaged which further effects the production.

Water bodies: Lakes and stream water become acidic and effects aquatic life. For example at < 5pH, fishes cannot hatch.

Buildings: It corrodes buildings and sculptures. For example, building made up of marbles are corroded.

Human : Through the acidification of soils, toxic metals such as mercury and aluminium is released. This ultimately goes to mix with drinking water, deposited in plants and fishes. Hence, human are affected by its consumption

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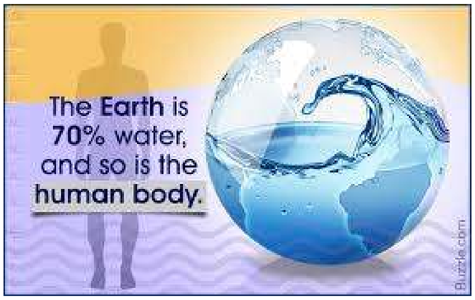

WATER POLLUTION

When toxic substances are present in water which harms plants and animals, this water is known as polluted water. There are mainly three types of indicators for water pollution:

- Physical pollution

It is because of color, pH, odor, conductivity.

- Chemical pollution

It is because of different chemicals dissolved in water, they are metals and non metals.

- Biological pollution

These are pathogens which causes diseases. For example: Bacteria, viruses, fungi etc.

Different forms of Water

- **Groundwater**

The rain water after falling on the earth's crust, some parts of it goes down as a leakage and stored as under groundwater . This water is used for drinking and other purpose extracted in the form of well, tubewell, boring etc.

- **Surface water**

This comprises river water, lakes, stream

- **Sea water**

All surface water system finally goes to sea water forming large water body which is saline water. Evaporation of this water causes seasonal rain.

Causes of water pollution:

1. Sewage: It is the wastes generated from household activities containing both grey and black water.It means the wastes contain urine, faeces, laundry waste, etc. These wastes may contain waterborne diseases. Diarrhea, typhoid, cholera, dysentery, etc., occur due to polluted water. Domestic sewage contains phosphorus and nitrates which causes unwanted algal bloom in ponds and lakes. These affect the water quality negatively.

2. Nutrients:These include pesticide, chemical fertilizer, manure, etc. Runoff from agricultural fields and farmland reaches to the nearby ponds, lakes streams, etc. These wastes can seep through the ground and cause ground water pollution as well as harm to aquatic life.

3. Waste Water: Waste water generated from household activities, hospitals, industries which contributes to pollute the water bodies after mixing, containing harmful chemicals.

4. Radioactive wastes: It's generated by uranium mining, nuclear power plants, and the production and testing of military weapons, as well as by universities and hospitals that use radioactive materials for research and medicine. Radioactive waste can persist in the environment for thousands of years, making disposal a major challenge.

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5. Oils pollution: The spillage of oils in sea by ships is the cause of oil pollution. It decreases DO of water thus effecting aquatic life there. Similarly, millions of cars and vehicles contributing to mix oils and gasolines into water body which ultimately goes to sea water.

6. Harmful chemicals: These are wastewater generated through industrial processing and hospitals. Wastes from iron and steel industry, food industry, chemical industry etc., pollute the water. These can contain toxic materials like heavy metals. When the waste stream mix with the natural water bodies, they become harmful to human and other organisms.

Effects:
Water is a universal solvent. Due to this nature of water, most of the substances are dissolved in water and easily get contaminated. This contaminated water when reaches to us, causes illness.

1. Diseases: Drinking polluted water causes various water born diseases. For example: diarrhea, dysentery, cholera, typhoid etc. which is one of the major causes of killer in the developing countries.

2. Ecosystems: It affects the ecosystem if left unchecked
3. Eutrophication: Chemicals in water body, for example, nitrogen and phosphorus from waste water encourage the growth of algae which forms a layer on top of the lake and ponds. Bacteria feeds this algae and dissolved oxygen in water goes decreasing so that it is difficult for aquatic life to survive
4. Food chain: When toxins and pollutants in the water are consumed by fishes and other aquatic animals, which are the food of human beings, after consumption, toxic substances reaches to us , causing illness

Control measures:

- Wastewater treatment and recycle and reuse should be encouraged
- Do not mix wastewater directly into river without treatment
- Create awareness to ensure that people understand the adverse impact of water pollution
- Follow proper methods to protect the rivers, lakes and seas and ensure that no wastes/. debris are dumped to prevent contamination
- Educate the people to create a culture of responsibility to reduce the disposal of wastes
- Avoid throwing recycled materials on water/river and keep the marine surroundings clean.
- Proper dumpings and management of solid wastes

Some indicators for water pollutions:

1. Alkalinity
2. Hardness
3. Free chlorine
4. Dissolved Oxygen(DO)
5. Biochemical Oxygen Demand(BOD)
6. Chemical Oxygen Demand(COD)

(Note: Students are encouraged to refer the Chemistry practical manual published by Nepal Engineering College for the above section)

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SOIL POLLUTION

Soil is the natural substance produced gradually due to weathering effects of rocks after thousands of years on the earth's crust. This is a home for different microbes and macrobes where plants grow and animals live. For various reasons it has been polluting. Anything that causes contamination of soils and degrade its quality is known as soil pollution. Soil pollution is nowadays major concern to the scientist because it is mainly caused by human activities in the name of development. Adding pollutants and thus degrading it.

Causes of soil pollution:

Pesticides: Use of pesticides in killing different rodents and insects produces harmful chemicals. In one hand they kill the animals, at long run, they effects the soil . For example DDT is used for pest control which produces Aldrin and Dialdrin when goes to soil, which is non biodegradable, slowly moves from one place to another. Ultimately, it reaches to human body, having serious health effect. Because of this effect of DDT, it has been banned nowadays.

Industrial wastes: The industrial wastes are if not managed properly, the toxic substances goes into the soil altering its quality. It may causes too acidic or too basic nature of soil. Thus it affects the soil fertility and increases the cost to recover the quality for suitable farming.

Acid rain: Acid rain helps to make the soil acidic and hence decrease its fertility.

Deforestation: Roots of the trees are helpful to bind the soils so that it protects the soil from erosion. Therefore, if trees are cut, there is high risk of soil erosion, which causes to degrade the soil

Barelanding:If the land is left bare, wind can easily carry soils from one place to another and hence degrading its quality. Therefore, plantation is needed.

Solid wastes: Due to mismanagement of solid waste, harmful chemicals are mixed into the soil. Disposal of plastics, cans, and other solid waste falls into the category of soil pollution. Disposal of electrical goods such as batteries causes an adverse effect on the soil due to the presence of harmful chemicals. For instance, lithium present in batteries can cause leaching of soil.

Rapid Urbanization:Lack of proper waste disposal, haphazard constructions can cause damage to the soil due to lack of proper drainage and surface run-off. It contains chemical waste from residential areas. Moreover leaking of sewerage system can also affect soil quality and cause soil pollution by changing the chemical composition of the soil.

Effects

Human health: With the exposure to harmful chemicals and toxic substances in the soil, it either deposited to plant parts or mixes with water bodies. When it is consumed, it causes illness.

Agriculture: It alters the soil character, making it acidic or sometimes more basic. This causes decrease in food production

Aquatic life: Harmful chemical contaminated soil if mixes with water, either groundwater or lake and pond water, river water, it ultimately affects the aquatic life. For example, if pH <5, fishes in the water cannot survive.

Ecosystem: Due to mixing of toxic chemicals, it destroys the soil, water and life associated with it. Therefore, whole ecosystem is affected.

Water source contamination: The contaminated soil once mixed with water bodies, it makes it polluted. For example, mismanaged solid waste causes nitrate contamination of groundwater. Likewise, heavy metals can be mixed with water which are present in the wastes.

Control Measures

- Practice of organic fertilisers and discourage to use chemical fertilisers
- Proper solid waste management practices
- Afforestation and discourage of deforestation
- Do not leave bare land
- Crop rotation practices while farming

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- Banned pesticide and harmful chemicals uses
- Industrial wastes should be managed properly
- Hospital wastes should be managed properly
- Awareness raising to all age groups
- Proactive plans and policies formulation by the government to preserve soil and all natural resources

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QUESTIONS

1. What are the different segments of environment?
2. What are the causes of environmental pollution
3. Write the different steps for the control of air pollution
4. What are the effects of air pollution
5. What do you mean by urbanization and its impact to natural resources?
6. What do you mean by acid rain? Why it happens? Write the reactions involved
7. Who are responsible for water pollution? Justify it.
8. What should be the action taken by government level& public level to control water pollution?
9. Why drinking water quality and quantity is a big challenge for all?
10. What are the different indicators for water pollution?
11. How alkaninity can be measured in the laboratory?
12. What do you mean by total hardnes and permanent hardnes of water? How it can be measured in the laboratory?
13. What are the proper ways for solid waste management taking the example of your community?
14. How mismanagement of solidwaste causes water pollution and soil pollution?
15. Discuss the ways to protect and increase the soil fertility by the farmers level, and by the policy level.
16. Food security is directly assocated with soil, water and air pollution, explain it.

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